



Privacy-Preserving Online Task Assignment in Spatial Crowdsourcing with Untrusted Server

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[3] Emory University

Outlines



- Introduction & Motivation
- Related work
- Background
- Proposed Approach
- Evaluation
- Conclusions



Crowdsourcing: outsourcing a set of tasks to a set of workers

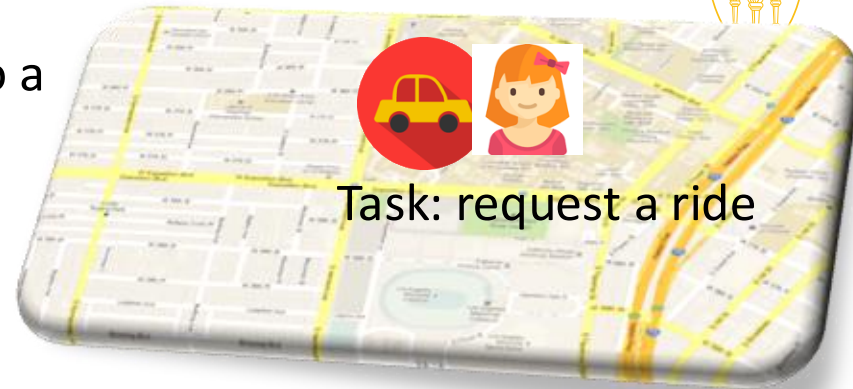




Crowdsourcing: outsourcing a set of tasks to a set of workers



Spatial crowdsourcing (SC): requires workers to *physically* travel to task's location

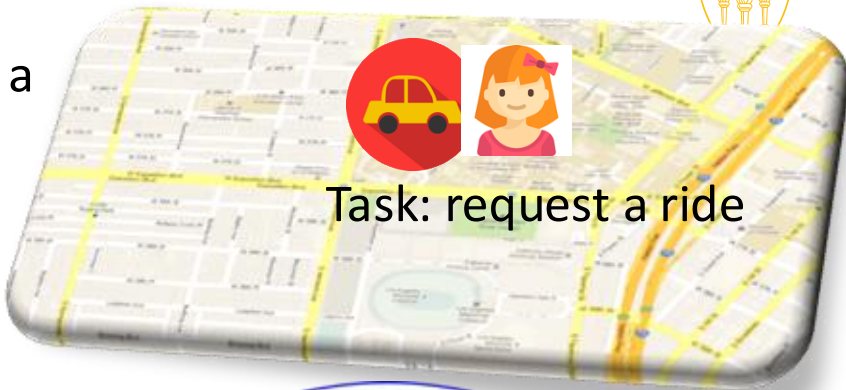




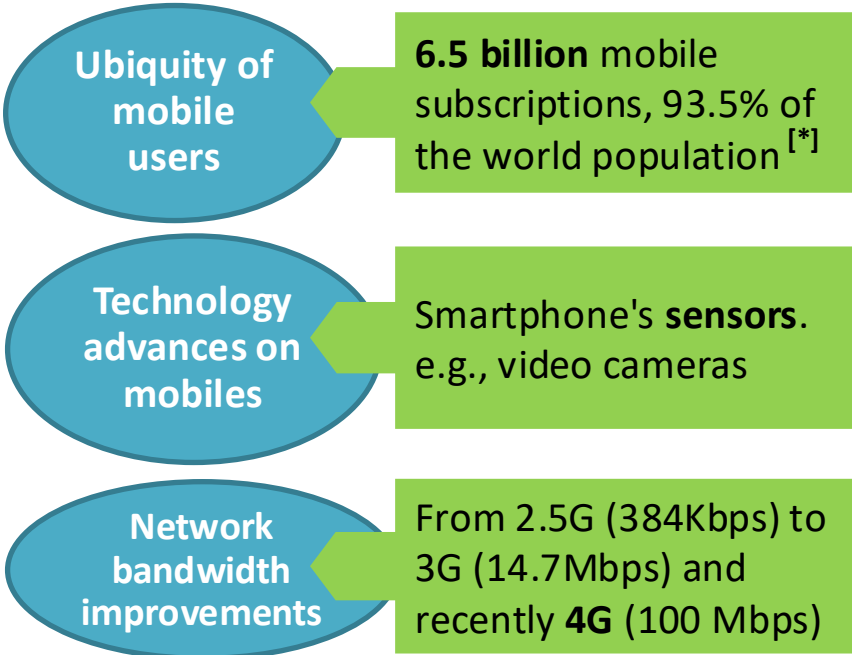
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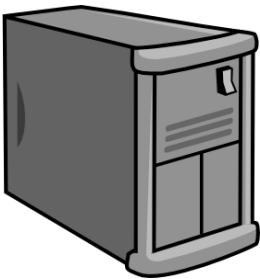
Task: request a ride



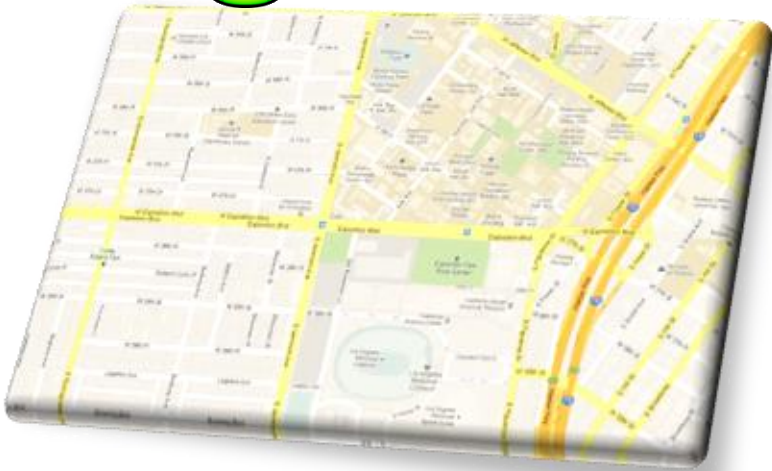
Task Assignment in SC



Requesters
(e.g., request
a ride)



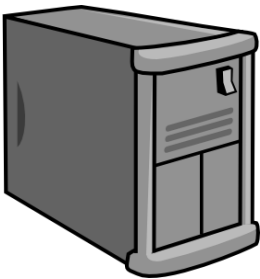
Server
(e.g., Uber)



Workers
(e.g., drivers)



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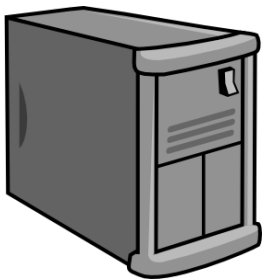
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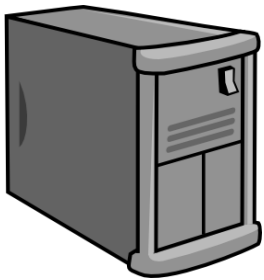


Workers
(e.g., drivers)

Server chooses best workers for a task based on task-worker proximity *e.g., [Kazemi'12, Pournajaf '14, To'17]*



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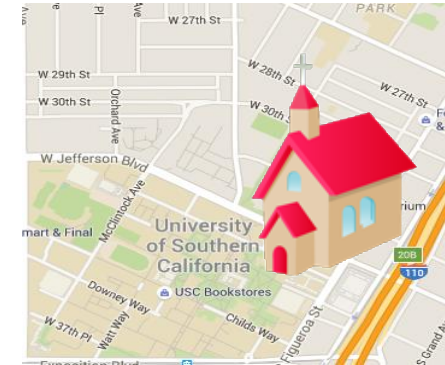
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Server knows locations of workers and tasks ☹️

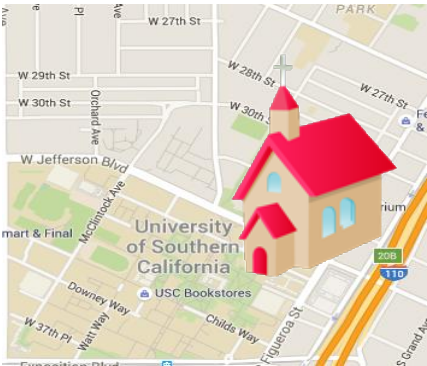


Location leaks sensitive information,
e.g., religious view, health status





Location leaks sensitive information, e.g., religious view, health status



Attacks based on locations:

PRIVACY ROAD KILL 4/26/16 2:40 PM

If you use Waze, hackers can stalk you

'God View': Uber Allegedly Stalked Users

“Uber treated guests to Creepy Stalker View, showing them the whereabouts and movements of 30 Uber users in New York in real time.”



Forbes





Anonymity based (e.g., cloaking)

- Pseudonymity [*Pfitzmann et al. 2010*]
- K-anonymity/Cloaking [*Sweeney'02*]

Encryption-Based

- Private information retrieval [*Ghinita et al. SIGMOD 2008*]
- Space transformation [*Khoshgozaran & Shahabi SSTD 2007*]

Perturbation (e.g., differential privacy)

- Geo-indistinguishability [*Andrés et al CCS 2013*]
- δ -location set-based differential privacy [*Xiao & Xiong CCS 2015*]



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Apple and Google adapted **differential privacy** to discover usage patterns from a large number of users

- Google Chrome web browser^[1]
- Apple QuickType/Emoji^[2] suggestions.





Papers	Privacy Techniques			Protection		Trusted Server	
	<i>Cloak</i>	<i>Encrypt</i>	<i>Perturb</i>	<i>Worker</i>	<i>Task</i>	<i>Yes</i>	<i>No</i>
[Pournajaf et al. 2014]	x			x		x	
[Sun et al. 2017]	x			x		x	
[Pham et al. 2017]	x			x	x	x	
[Hu et al. 2015]	x			x		x	
[Shen et al. 2016]		x		x			x
[Liu et al. 2017]		x		x	x		x
[To et al. 2014]			x	x		x	
[Gong et al. 2015]			x	x		x	
[Zhang et al. 2015]			x	x		x	
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Existing work that use perturbation technique protect worker location only and assume trusted server ☹️

Outlines

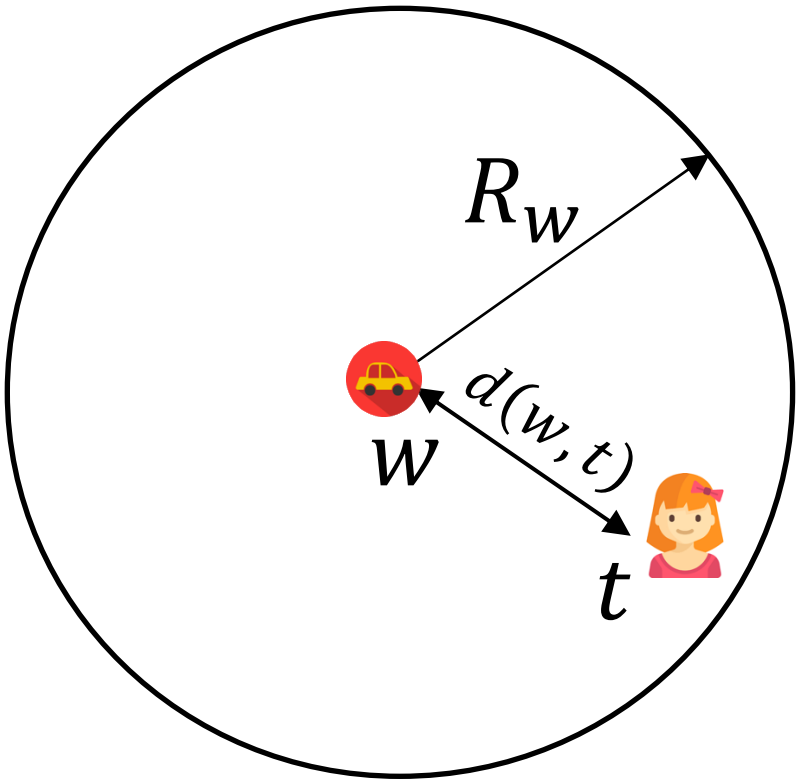


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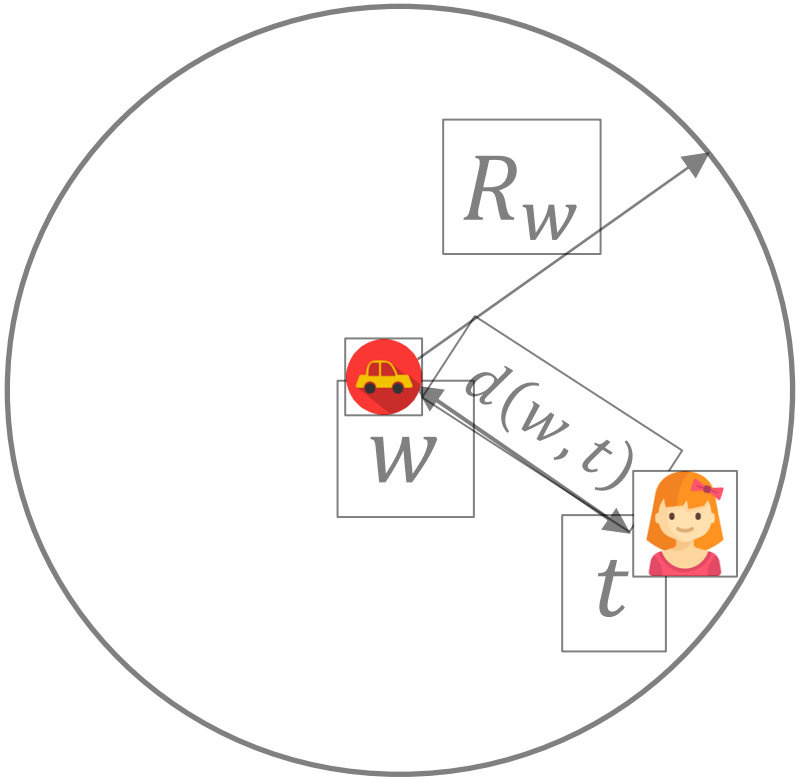


Notation	Description
w, t	Actual locations of a worker, a task
w', t'	Perturbed locations
R_w	Reachable distance of worker w
$d(w, t)$	Euclidean distance between w and t





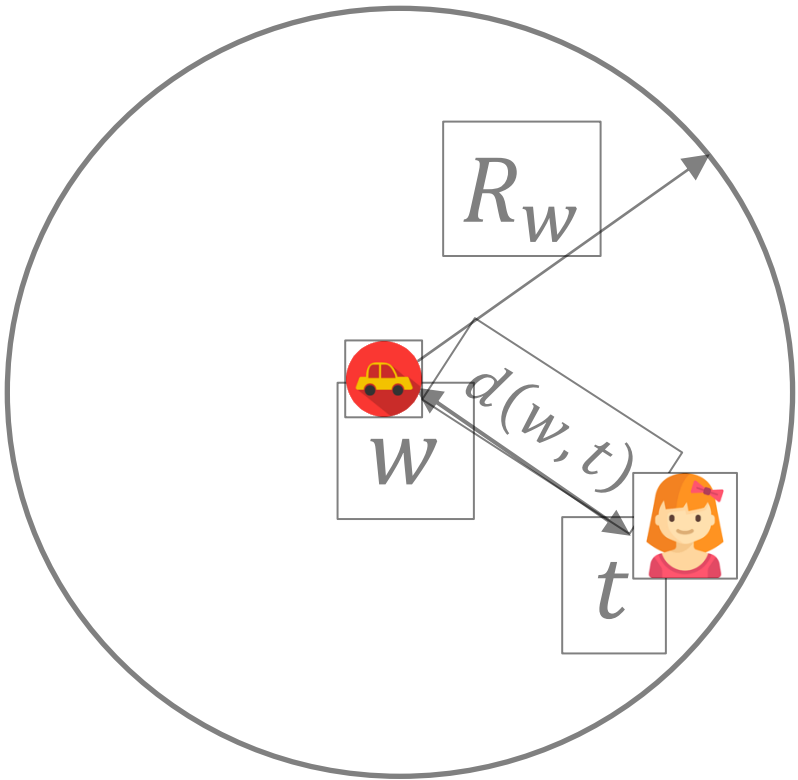
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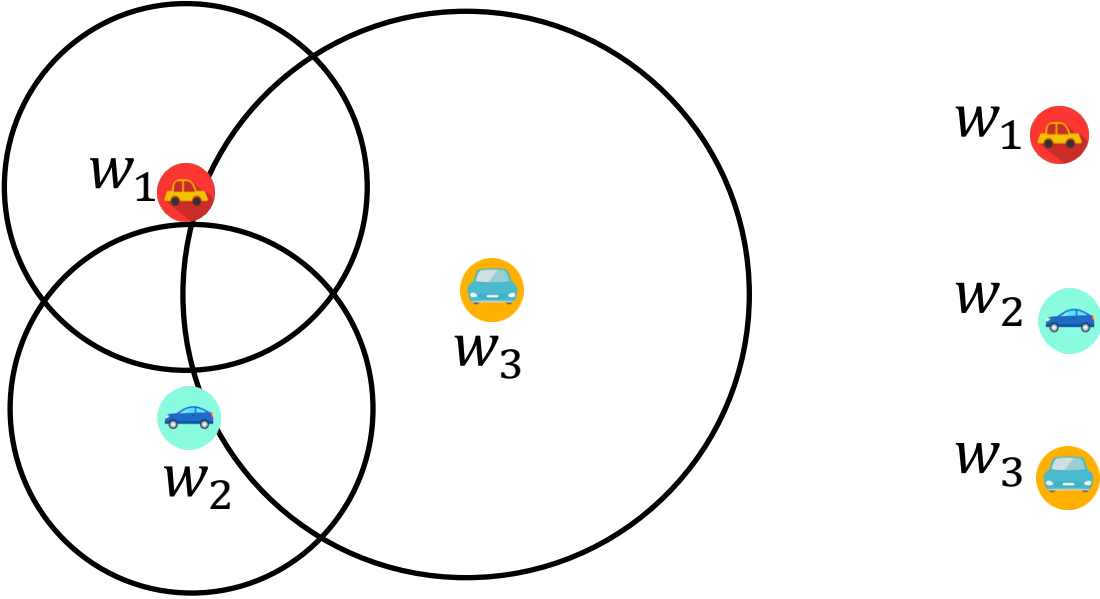
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d can be non-Euclidean & R_w can be complex shapes like polygon



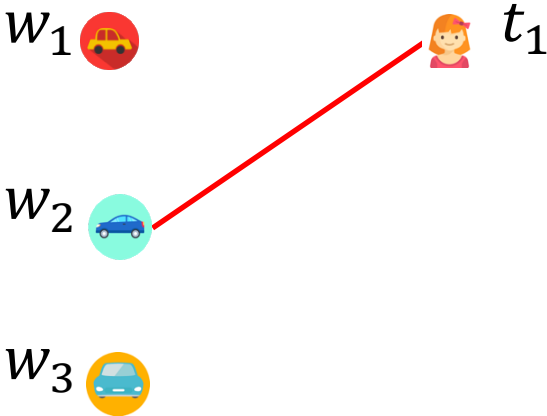
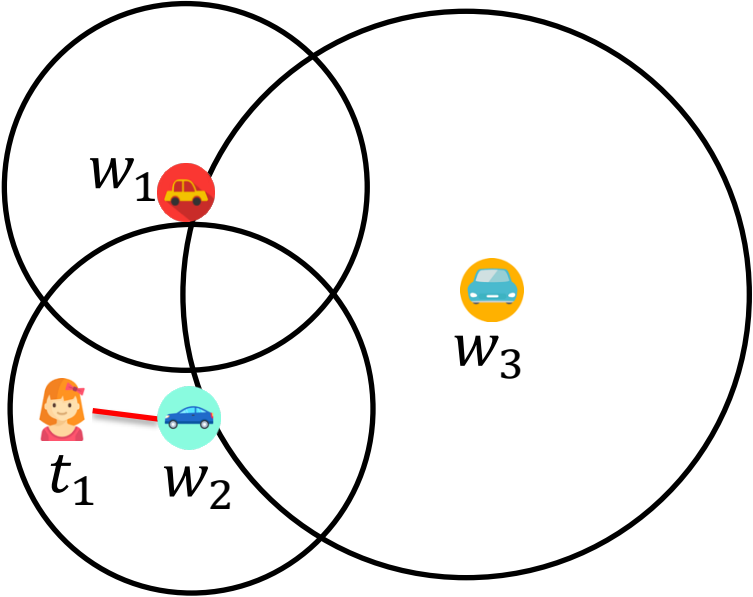


Worker set is known, each task arrives one-by-one



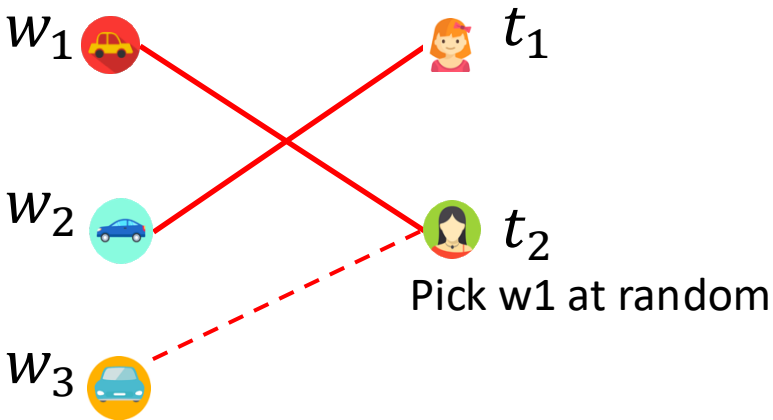
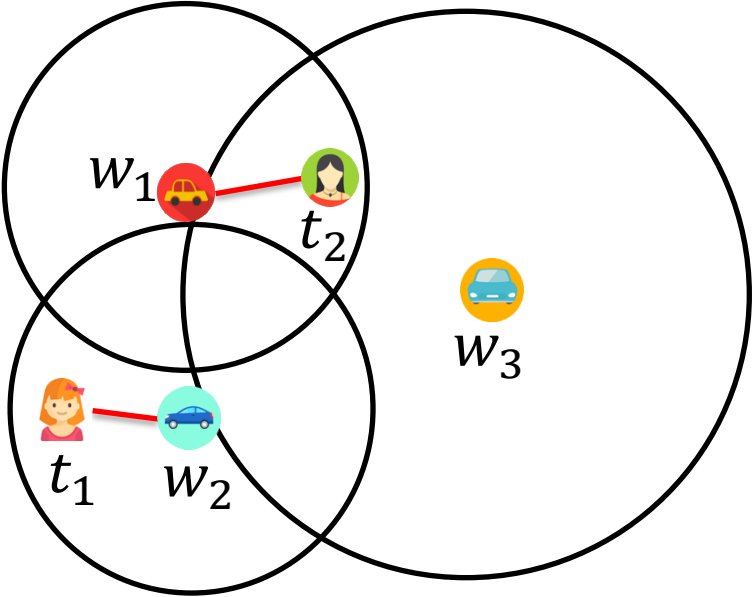


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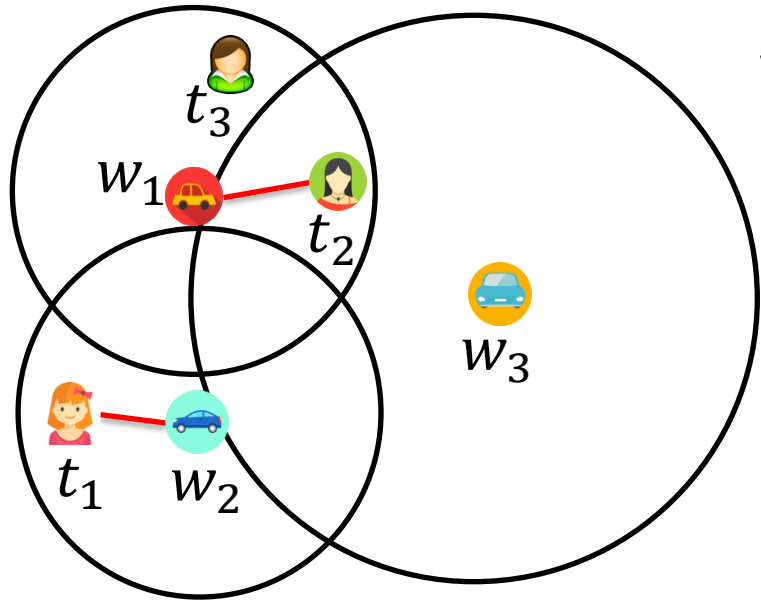


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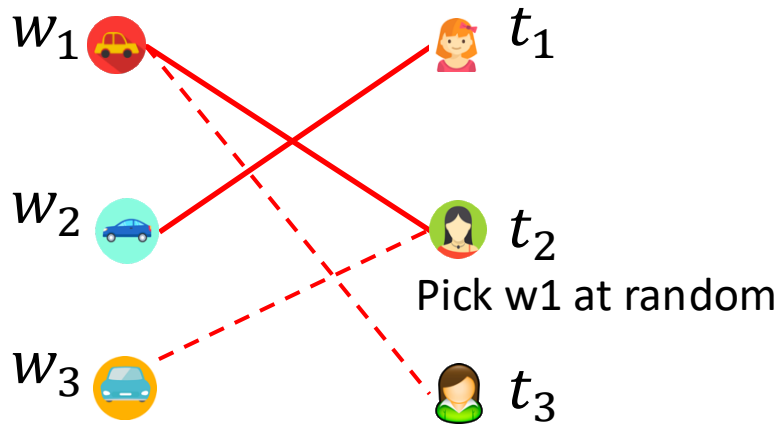




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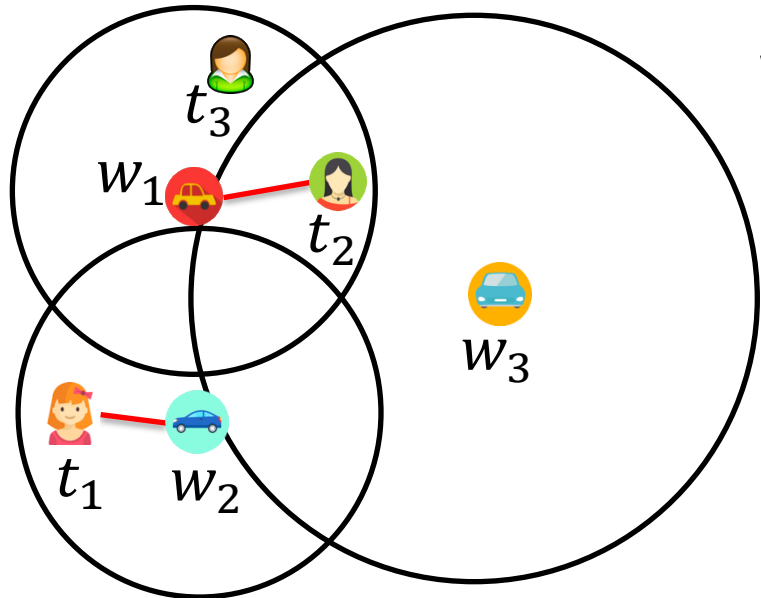


w1 is no longer not available

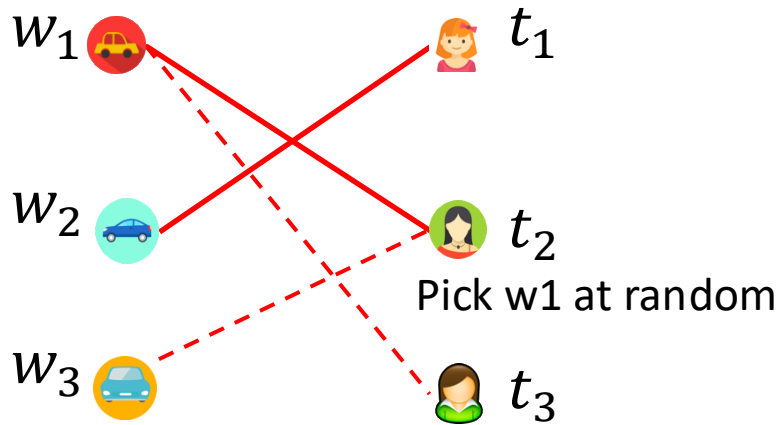




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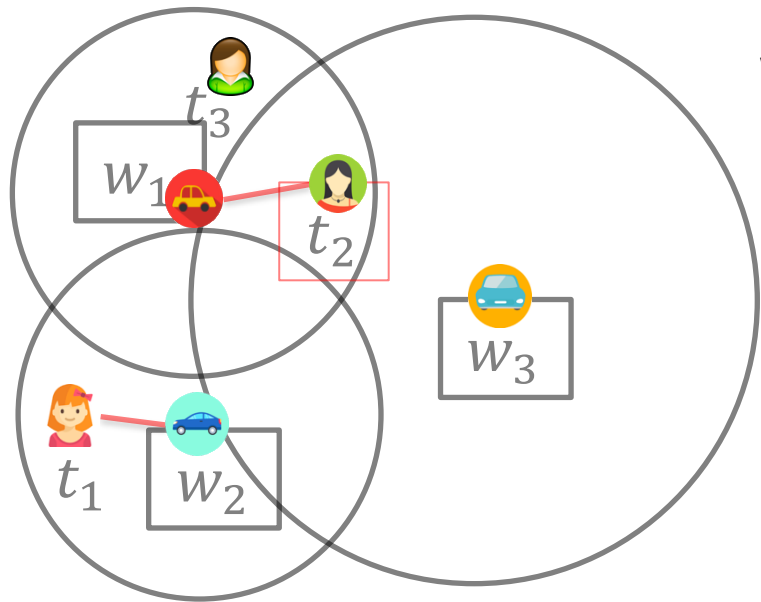
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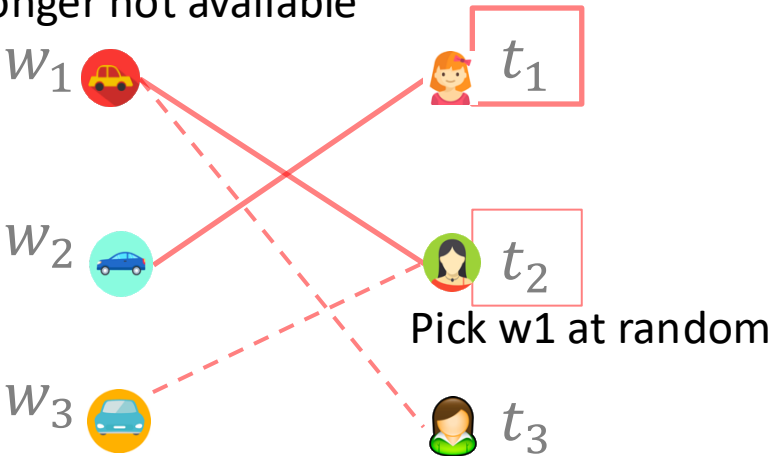
Assign as many tasks as possible to workers



Worker set is known, each task arrives one-by-one



w_1 is no longer not available



Assign as many tasks as possible to workers

Ranking algorithm^[*] is optimal, competitive ratio 0.63

- Permutes workers and assigns a **random rank** to them
- Each task is matched to a reachable worker of the highest rank

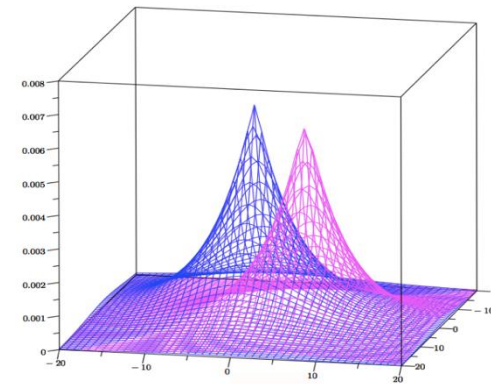
[*] Karp et al. *An optimal algorithm for on-line bipartite matching*, STC'90



The goal: An adversary cannot distinguish locations which are at most r distance away

Approach: Any two locations at distance at most r produce “similar” observations (bounded by ϵ),

- r is the radius of concern within which privacy is guaranteed (This means that an adversary cannot distinguish locations which are at most r distance away)
- The smaller ϵ is, the stronger privacy (as it gets harder for the attacker to detect the user's location among the points within this circle).



More formally:

Mechanism A satisfies (ϵ, r) -Geo-I iff for all x, y such that $d(x, y) \leq r$:

$$d_p(A(x), A(y)) \leq \epsilon d(x, y) \leq \epsilon r$$

• $d(x, y)$: Euclidean distance between x, y

• $d_p(.,.)$: multiplicative distance between two distributions

Why (ϵ, r) -Geo-I ?



“A mechanism satisfies ϵ -geo-indistinguishability iff for any radius $r > 0$, the user enjoys ϵr -privacy within r ”

- This definition requires that the user is **protected within any radius r , but with a level $l(r) = \epsilon r$ that increases with the distance.**
- (ϵ, r) -Geo-I means that we should **fix radius r** to a certain distance that users feel safe (say 1km) and then tune ϵ to balance privacy/utility trade-off. The smaller $l(r) = \epsilon r$ the higher privacy.
 - Given a small radius ($r=1$ km), $l(r)$ is small, guaranteeing that the attacker cannot infer the user's location within 1km
 - A large radius ($r=10.000$ km), $l(r)$ becomes very large, allowing the attacker to infer that with high probability the user is located in some place. Hence, more noise need to be added (smaller ϵ) to ensure similar privacy guarantee when $r=1$ km.



it is sufficient to achieve (ϵ, r) -Geo-I by generating random point z (from actual point $x \in X$) according to planar Laplace distribution.

r (in meters) is the radius within which privacy is guaranteed
 ϵ tunes how much privacy, smaller ϵ means higher privacy



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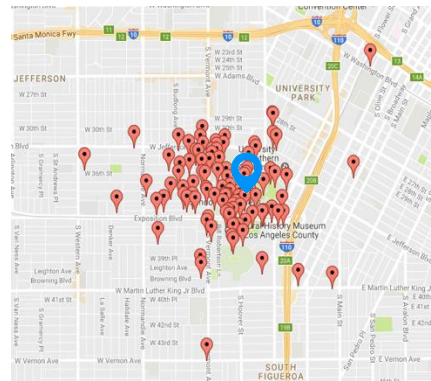
achieve privacy by injecting planar Laplace noise



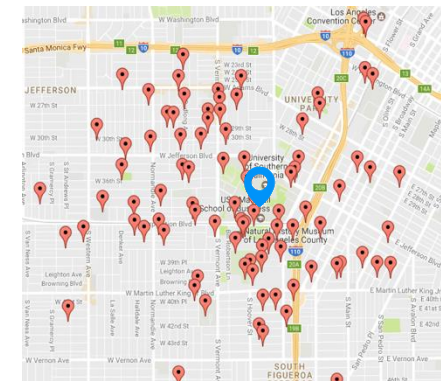
True locations



Perturbed locations



$\epsilon = \log(6)$
 $r = 1 \text{ km}$

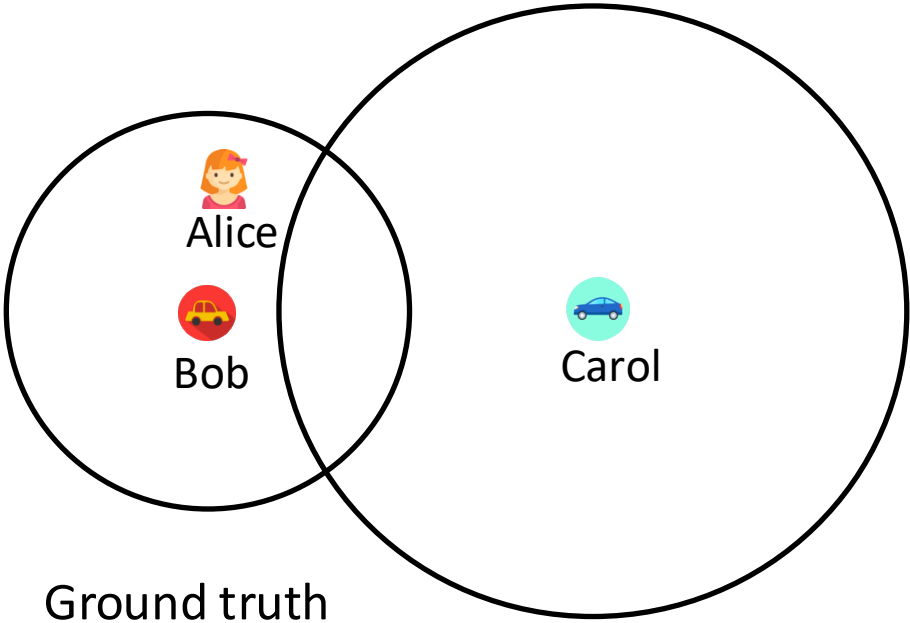


Better privacy: $\epsilon = \log(2)$
 $r = 1 \text{ km}$



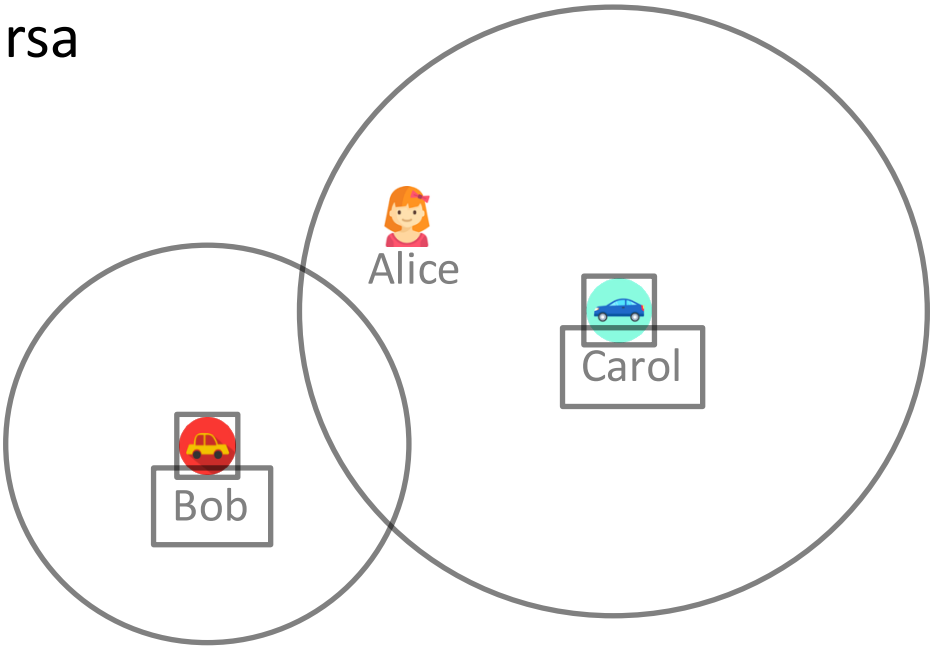


Reachable worker-task pair is observed as unreachable,
and vice versa





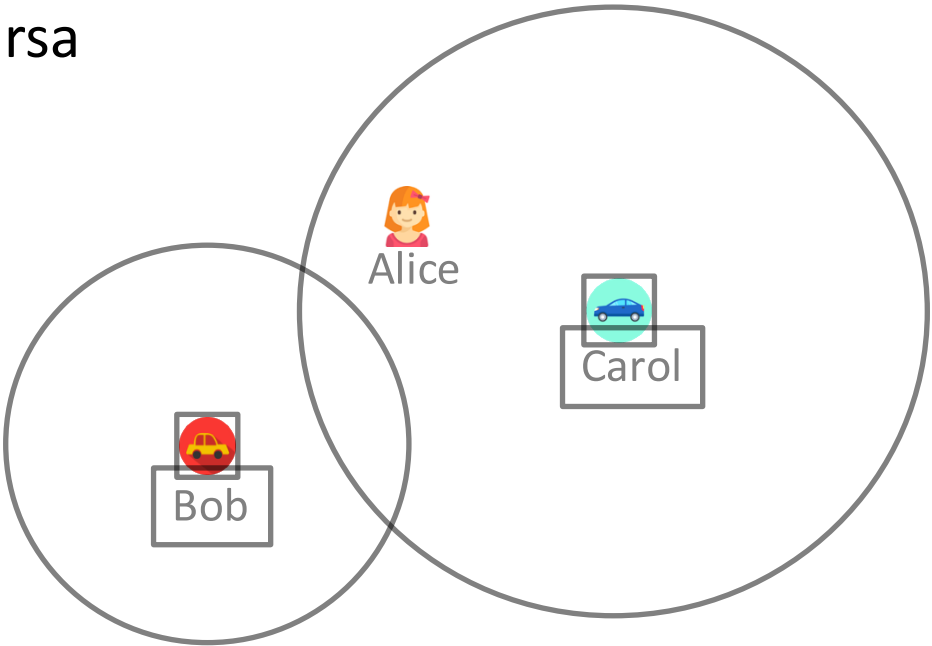
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They add noises locally to protect their true locations



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They add noises locally to protect their true locations

Alice is not assigned to Bob (not reachable) ☹️

Alice's location is disclosed to Carol *unnecessarily* ☹️

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Three-Phase Framework



Alice  requests  for a ride



Alice  requests  for a ride



Finds ***candidate drivers*** for Alice: Bob  Carol  Dave 

Server does not know *anyone's* location (works in perturbed space for both riders and drivers)

Sends perturbed locations of drivers to Alice



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Finds the **most likely reachable driver**: Bob 

Alice does not know *any driver's* location (works in perturbed space for drivers but knows her own location)

Reveals her location to Bob

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Reachable → accepts (happy case)

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Repeat until either task is assigned or no candidate worker left

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Location Disclosure

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System Overhead: size of the worker candidate set, captures communication and computational overhead

Location Disclosure (false hit): privacy leak occurs when Alice estimates an unreachable worker as reachable & reveals her location

Utility: number of assigned tasks

Worker Travel Cost: captures travel cost or assignment quality



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“Oblivious” algorithm

-   assumes perturbed locations as actual ones
- Direct adaptation of Ranking algorithm^[*] to our framework
 - Consider both **random rank** and **distance-based rank**



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Core idea:

-   to use underlying distributions of noisy locations to estimate real locations

Worker-Task Reachability





Compute the **reachability probability** of a worker-task pair given their observed distance



$$: \Pr(d(w, t) \leq R_w \mid d(w', t'))$$



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I. Analytical approach, based on estimating the reachability probability

II. Empirical approach, based on synthetic or historical data



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I. Analytical approach, based on estimating the reachability probability

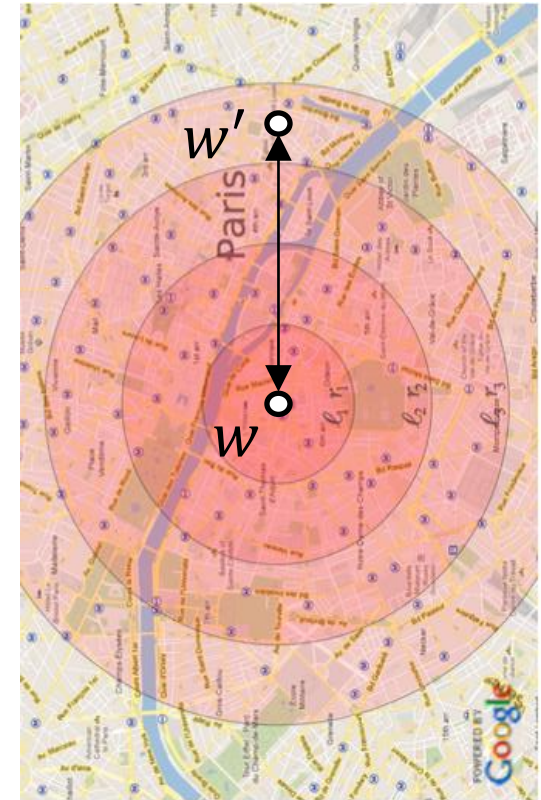
- Derive PDF of $d(w, t)$, given w', t'
Subsequently, the reachability probability can be computed efficiently
- Planar Laplace distribution is difficult to analyze so we approximate it by bivariate normal distribution (BND)

II. Empirical approach, based on synthetic or historical data





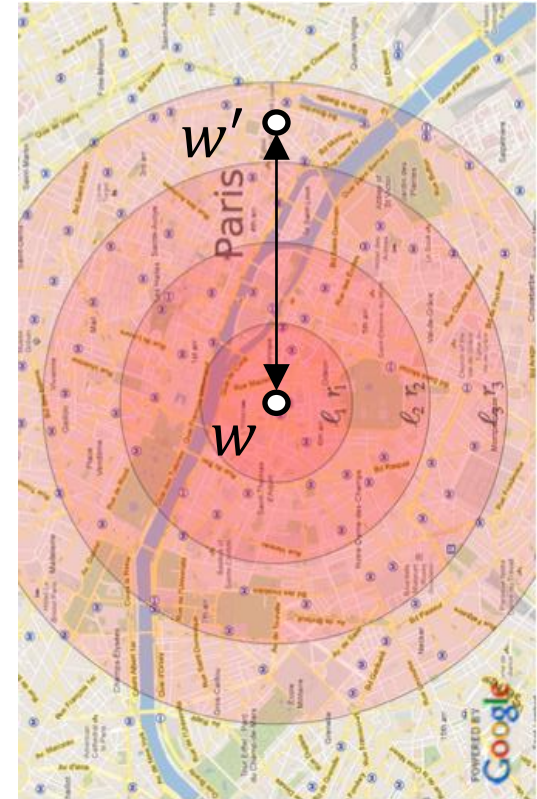
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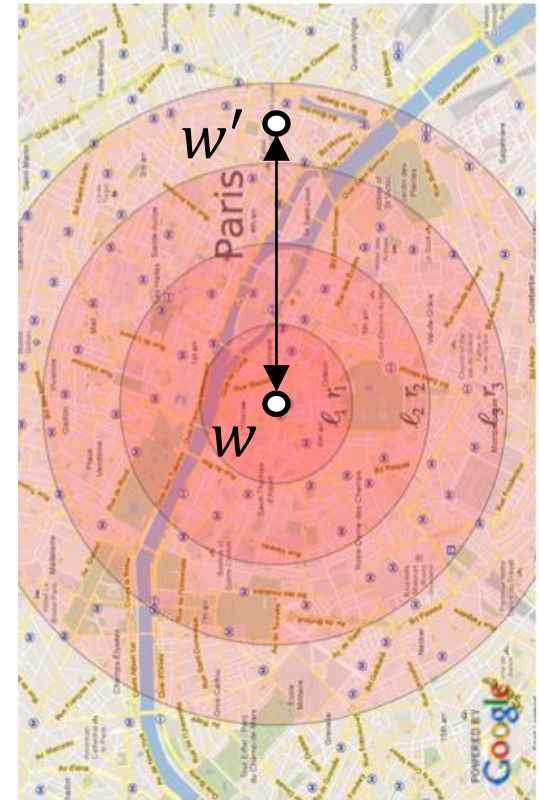
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- PLD is difficult to analyze

Approximate PLD by a circular BND with same

mean (w_x, w_y) & covariance matrix $\begin{bmatrix} \frac{2r^2}{\epsilon^2} & 0 \\ 0 & \frac{2r^2}{\epsilon^2} \end{bmatrix}$

- BND is made up of two random variables x and y ; both normally distributed
- PLD is symmetric to its center $\rightarrow$ approximated BND should be symmetric to the same center





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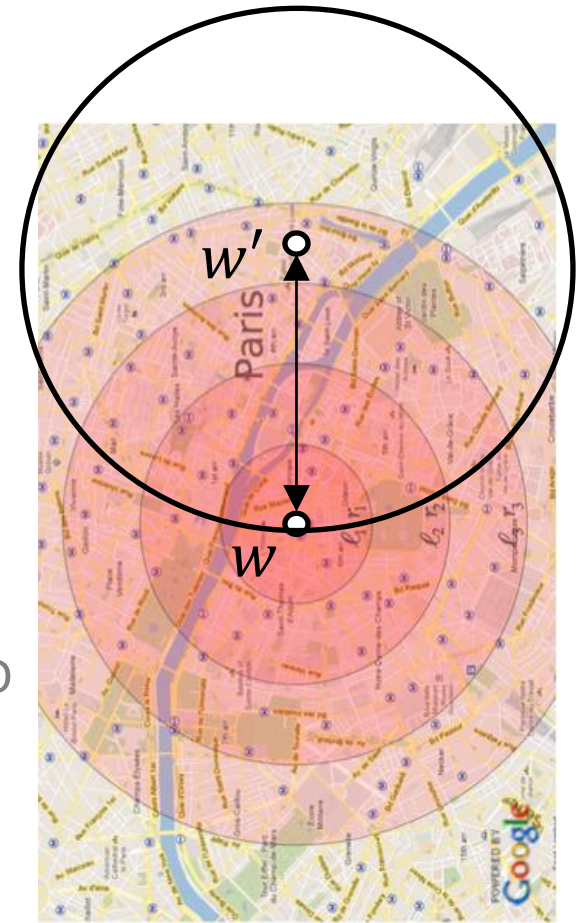
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w' is known $\rightarrow w$ follows circular BND centering at w' : circular $BND(w', \Sigma)$





derives PDF of $d(w, t)$





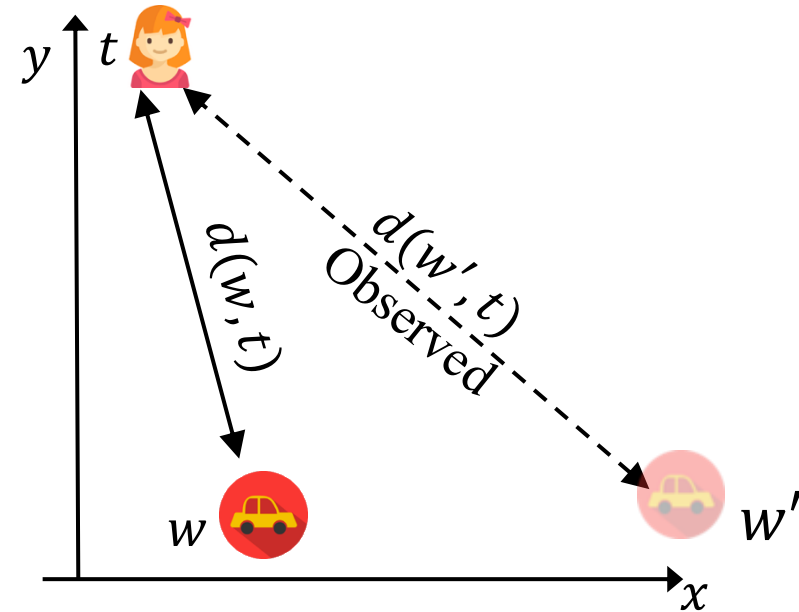
derives PDF of $d(w, t)$



Given true location of Alice  t and perturbed location of Bob  w'



estimates PDF of $d(w, t)$



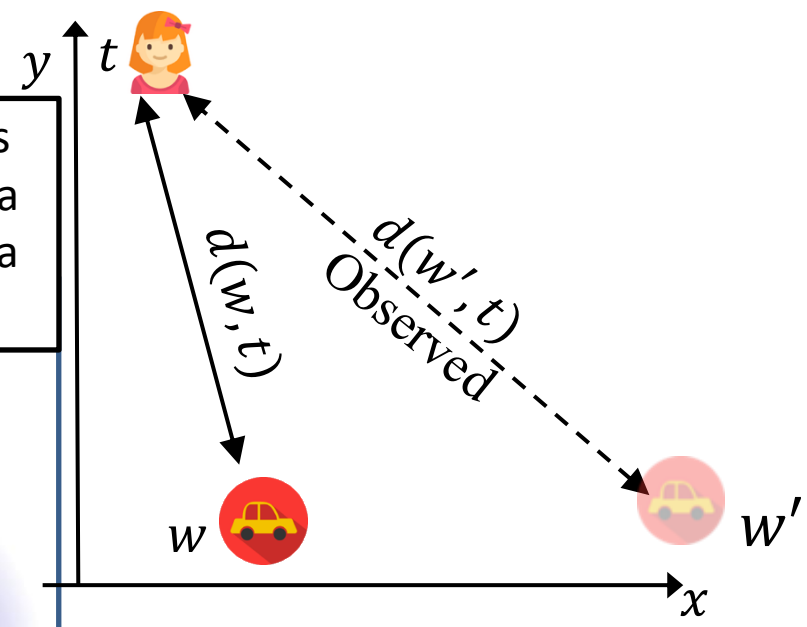
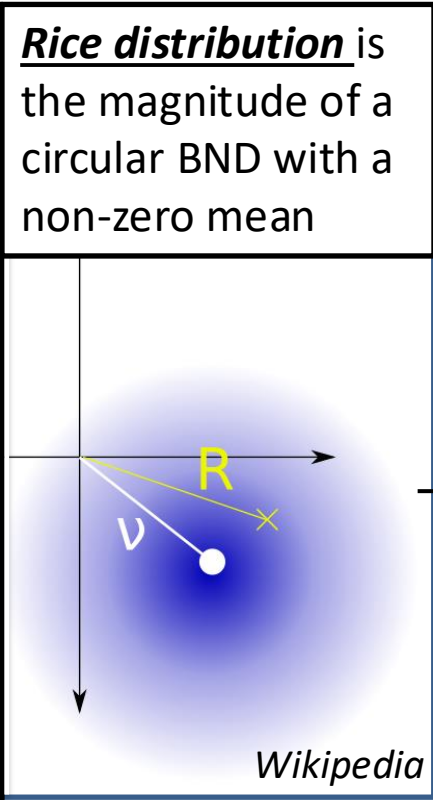


Given true location of Alice  t and perturbed location of Bob  w'



estimates PDF of $d(w, t)$

In the 2D plane, pick a fixed point at distance v from the origin. Generate a distribution of 2D points centered around that point, where the x and y coordinates are chosen independently from a [gaussian distribution](#) with standard deviation σ (blue region). If R is the distance from these points to the origin, then R has a Rice distribution.





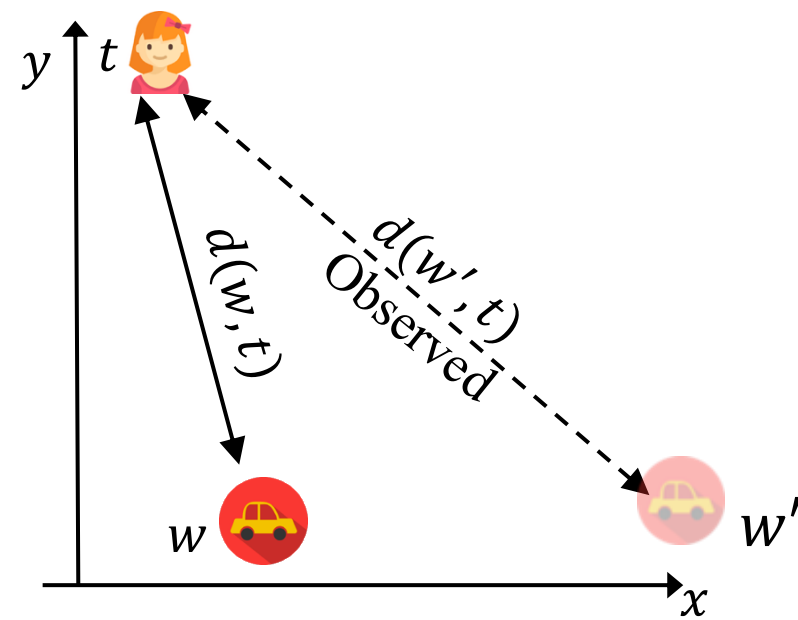
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estimates PDF of $d(w, t)$





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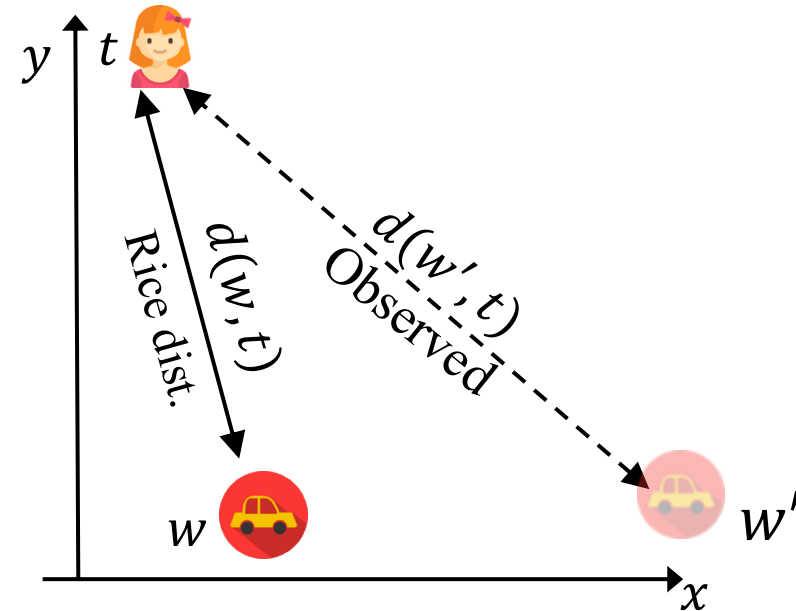
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- $\nu = d(w', t)$
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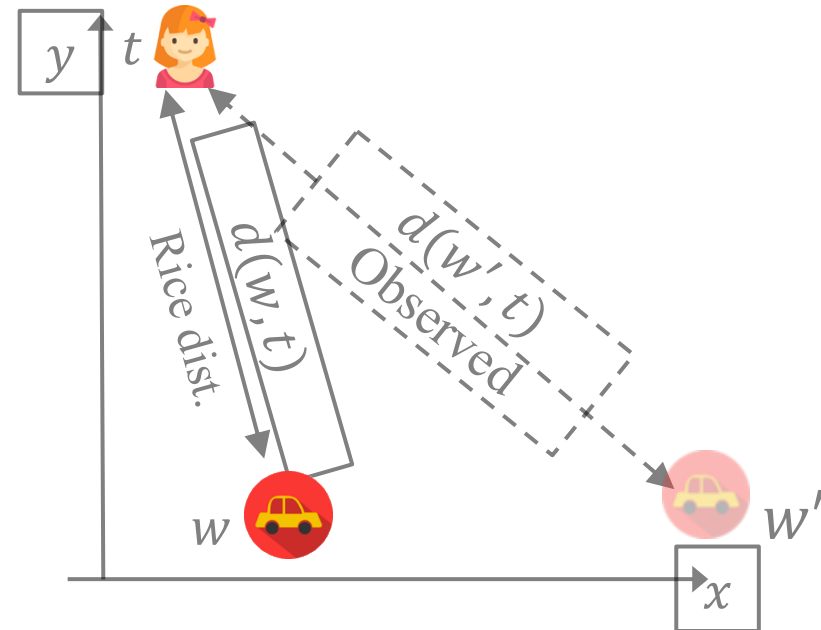
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How  estimates PDF of $d(w, t)$ can be found in the paper



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Heuristic:



can reduce disclosure of her location based on *reachability threshold* β ($\beta > \alpha$)

e.g., if $\text{Rank}_{w_i} < \beta$, cancel this task

Outlines



- Introduction & Motivation
- Related work
- Background
- Proposed Approach
- **Evaluation**
- Conclusions



- GPS-equipped taxis dataset ^[1]
 - Workers’ locations are the most recent drop-off locations
 - Tasks’ locations at the pick-up locations
 - 500 tasks and 500 workers were randomly sampled

	#Passengers	#Drivers	Area
T-Drive	100,000+	9,019	Beijing City

[1] Yuan et al. *T-drive: driving directions based on taxi trajectories*. SIGSPATIAL 2010



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- Performance metrics
 - **Utility**: number of assigned tasks
 - **Worker Travel Cost**: captures travel cost or assignment quality
 - **System Overhead**: size of the worker candidate set, captures communication and computational overhead
 - **Location Disclosure** (false hit): privacy leak occurs when requester estimates an unreachable worker as reachable

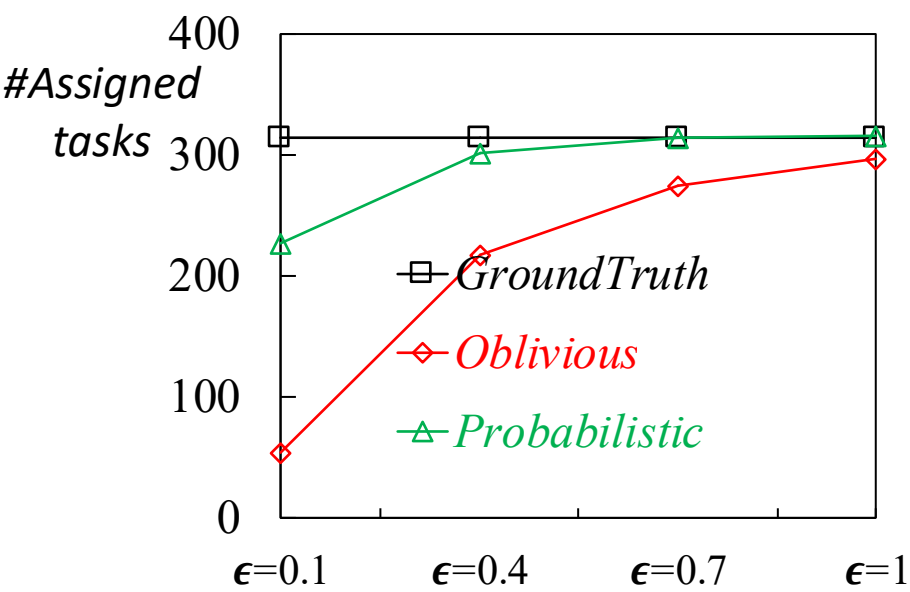
Utility and Travel Cost



<i>GroundTruth</i>	Has access to exact locations (distance-based rank)
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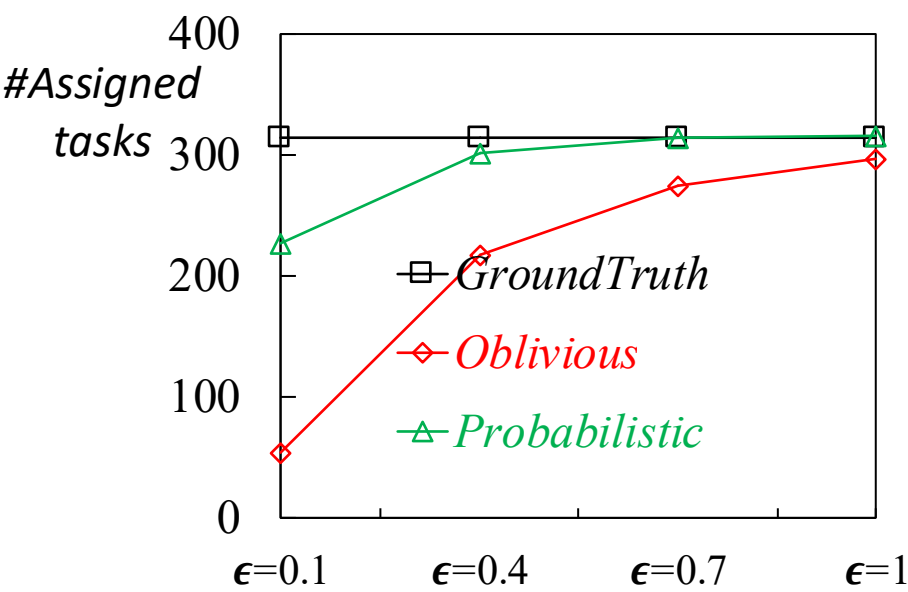


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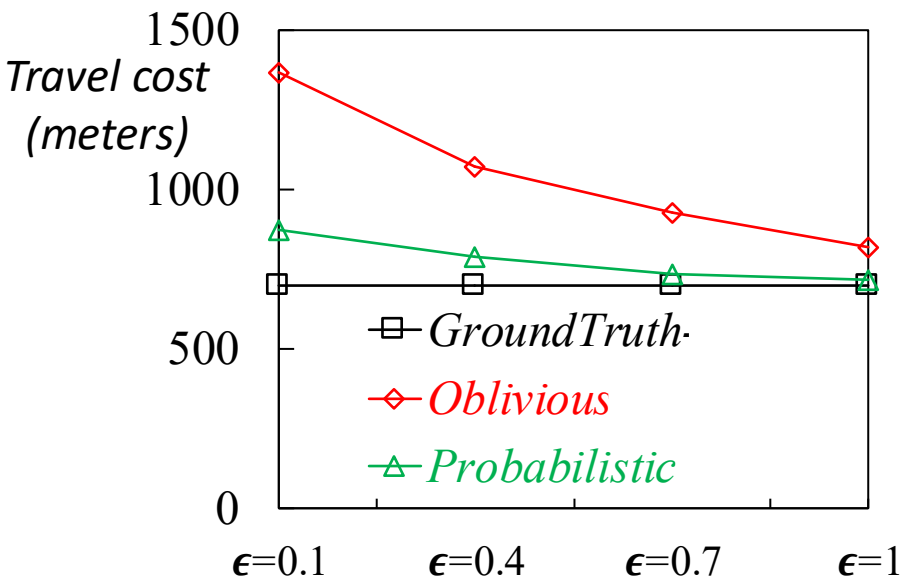
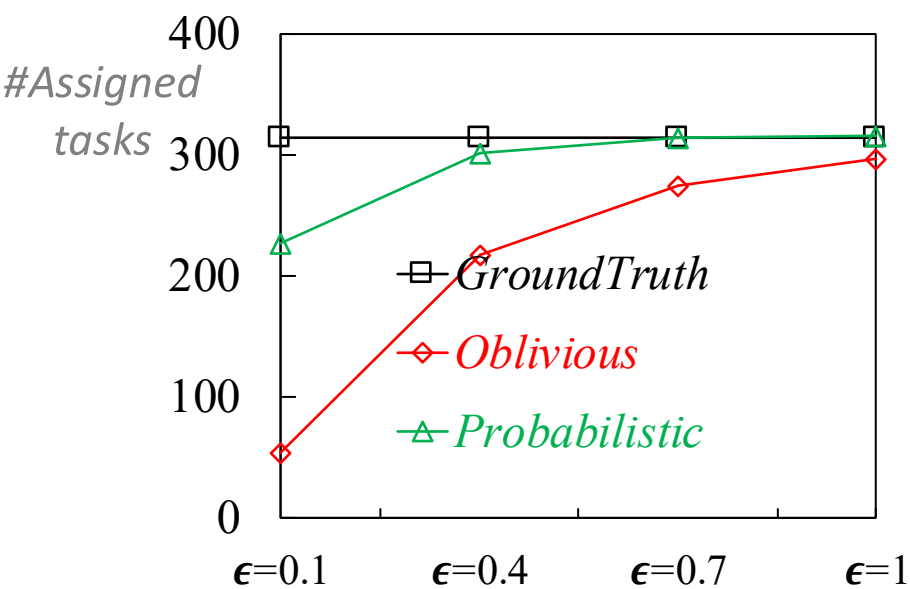
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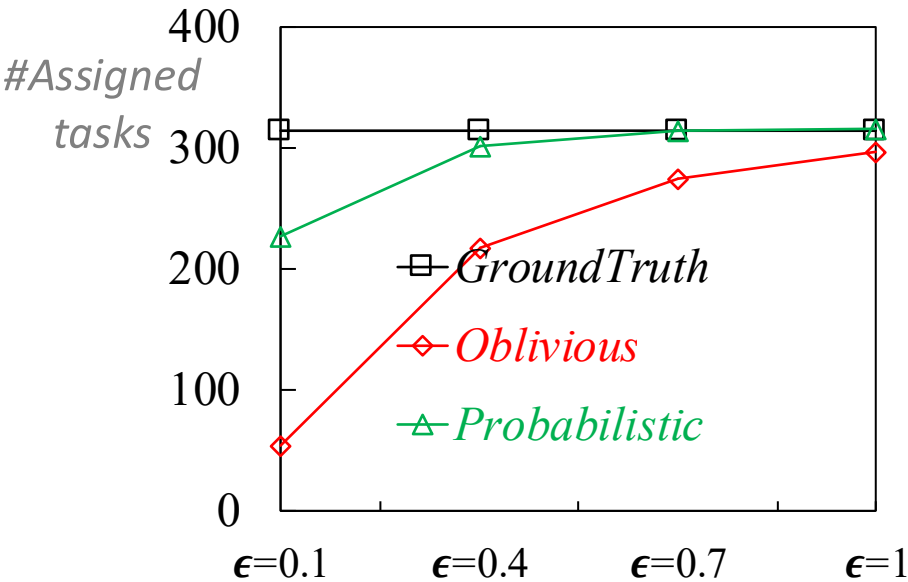
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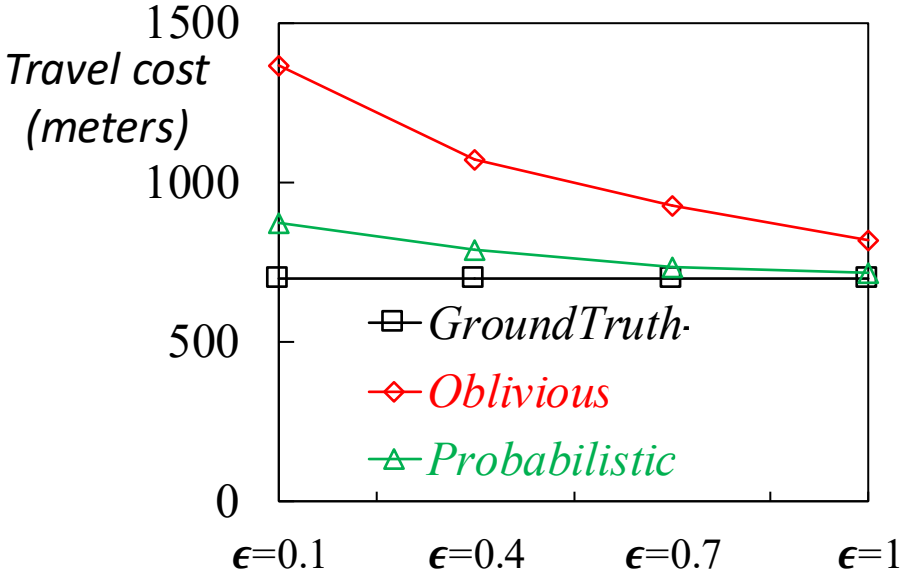
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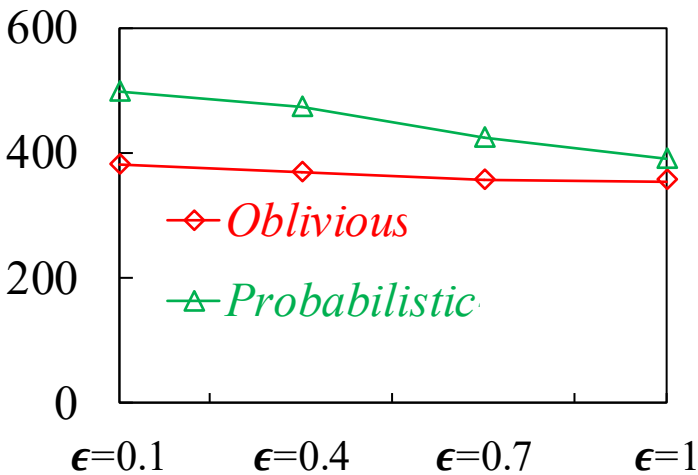


Probabilistic obtains significantly **lower travel cost** than *Oblivious* (by 30%)

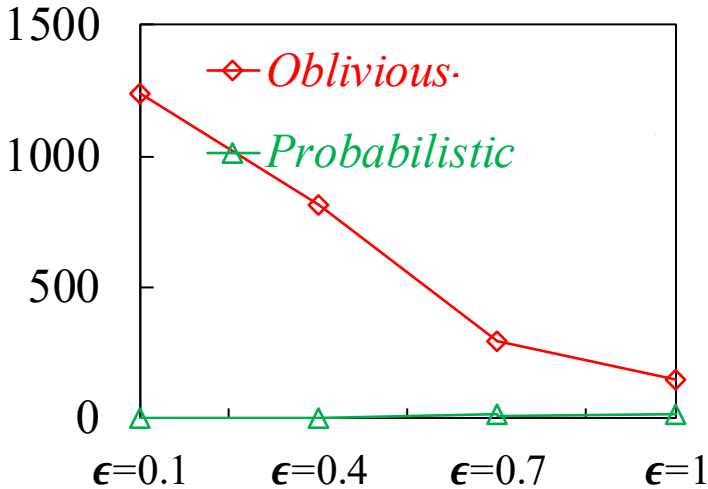


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#workers (overhead)



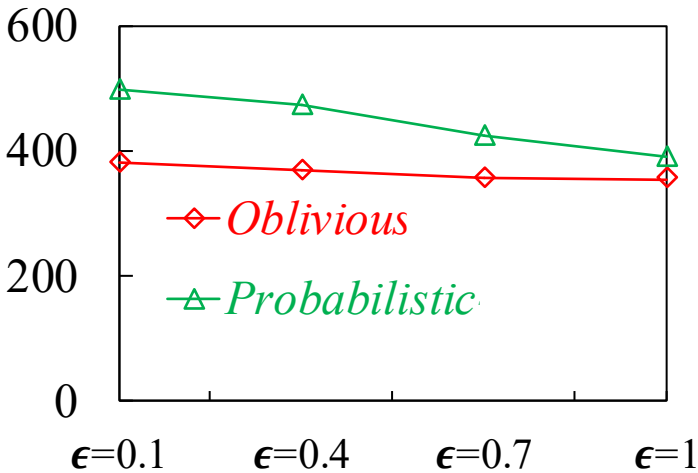
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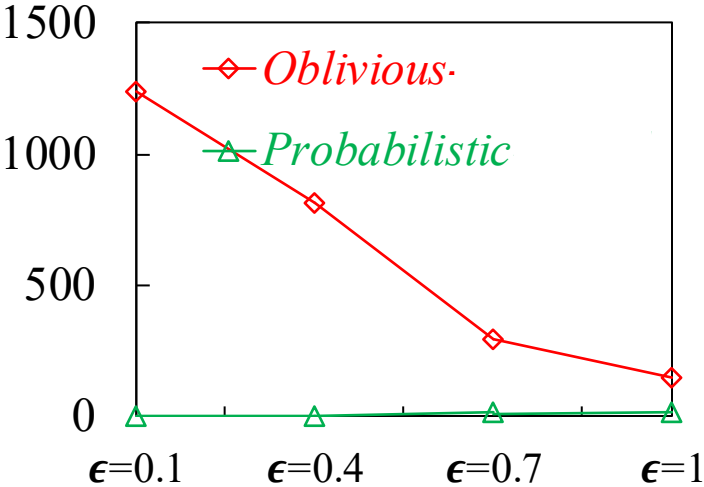


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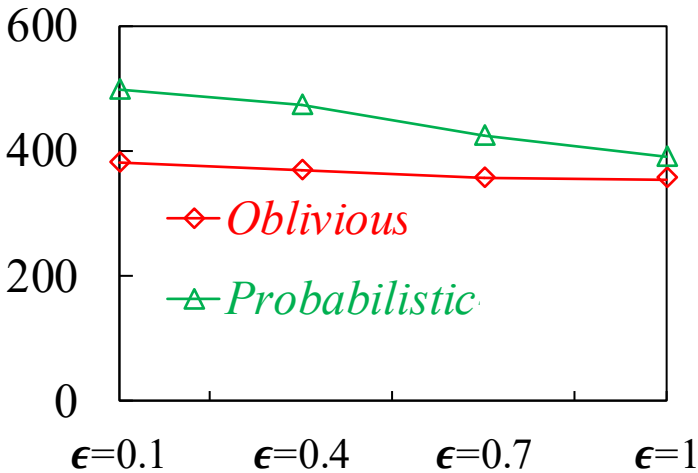


Although the overhead of *Probabilistic* is slightly higher than *Oblivious*'s, *Probabilistic* has **much smaller false hits**

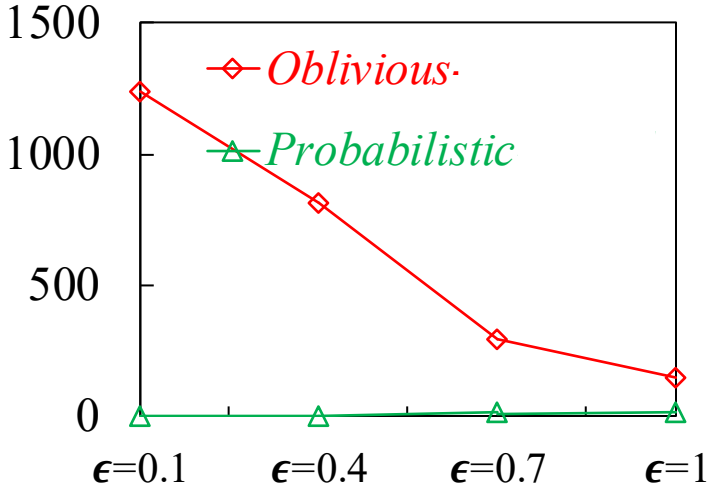


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Average **#false hits** before a task can be assigned: 23 workers vs 1.05 workers

Conclusions and Future Work



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 - Consider tasks that may require redundant assignment: taking pictures of a particular location, reporting how crowded a restaurant is



Two Sides of the Coin

*Protecting against
social inferences
* But allow for LBS*



Privacy Twist



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Inferring Social Relationships

- Privacy attack



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walk2friends: Inferring Social Links from Mobility Profiles
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- Can we do better in very dense datasets ?
- Feature learning method – Unsupervised
 - As opposed to EBM's supervised linear regression.
 - Claims to exploit followship in addition to EBM's co-occurrence
- Inspired by Deep Learning in NLP – word2vec
 - Skip-gram Model
(Tomas Mikolov et. al., at Google Research, 2013)



A glance at the Skip-Gram Model

Corpus training (NN)

The quick brown fox jumps over the lazy dog.



(fox, quick)
(fox, brown)
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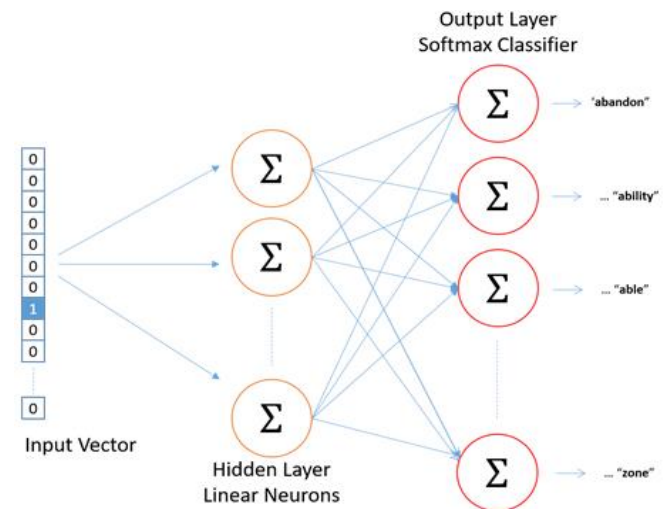
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Goal: Given a specific word in a sentence, tell us the probability for every word in our vocabulary of being the “nearby word” to the one we chose.

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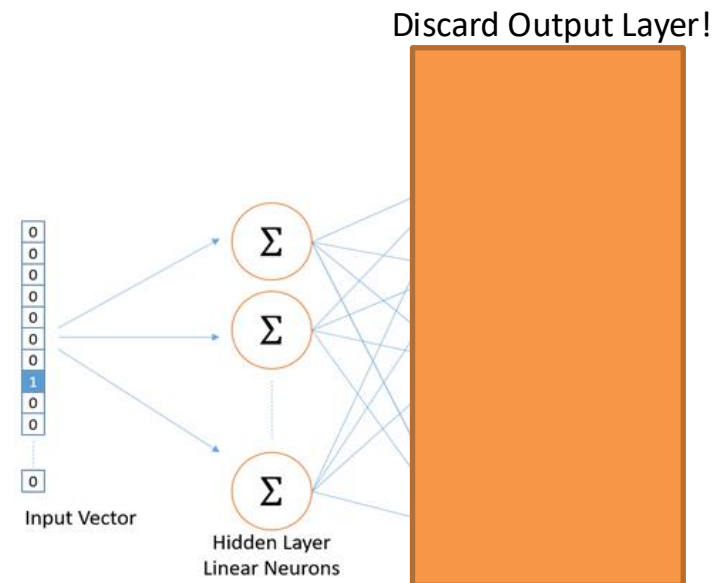
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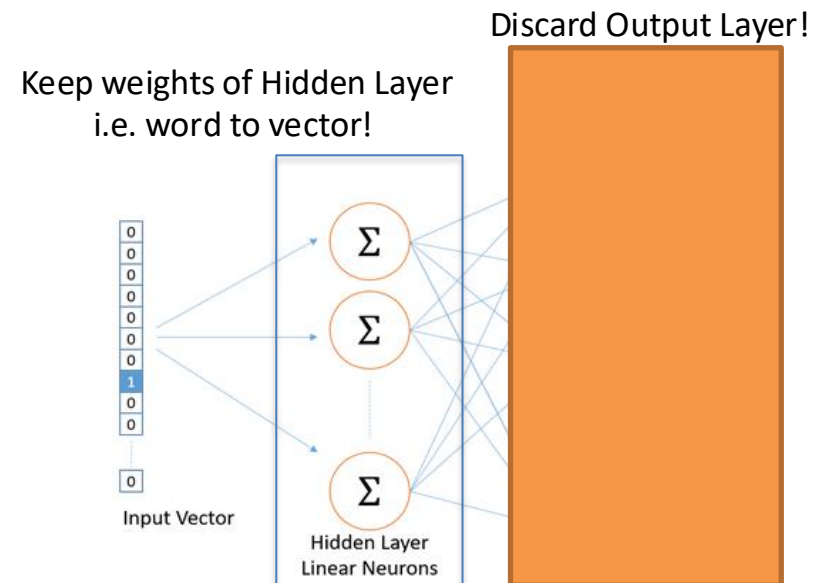
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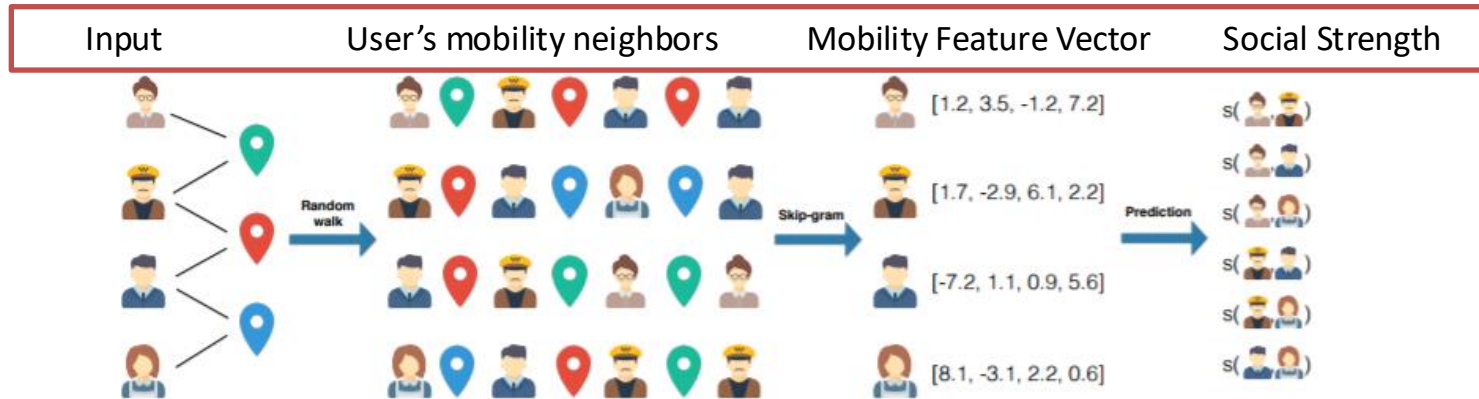
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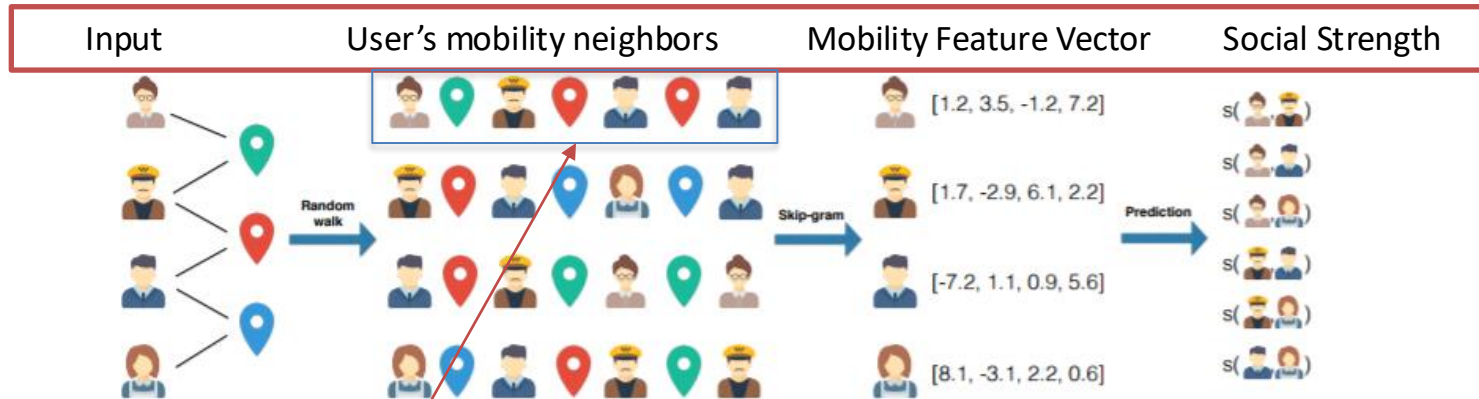


walk2friends: Extending to locations based networks.





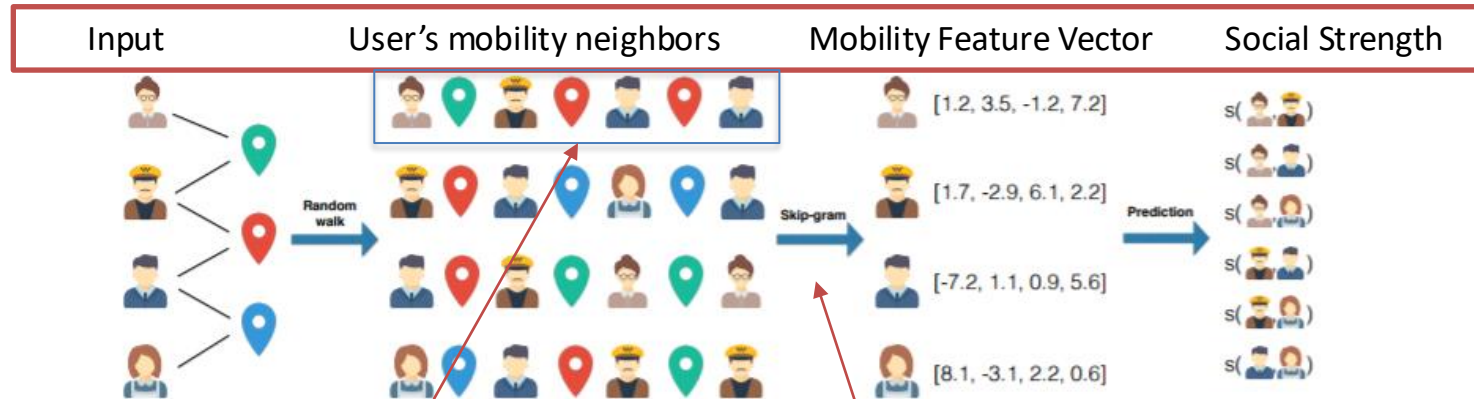
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Similar to corpus sentences



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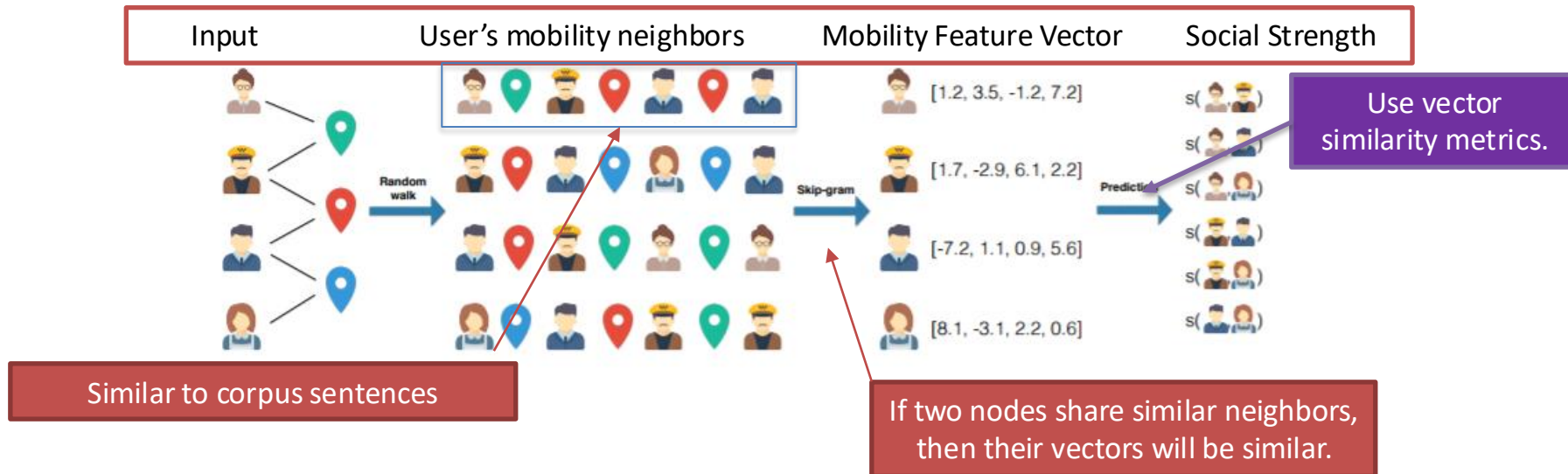


Similar to corpus sentences

If two nodes share similar neighbors, then their vectors will be similar.

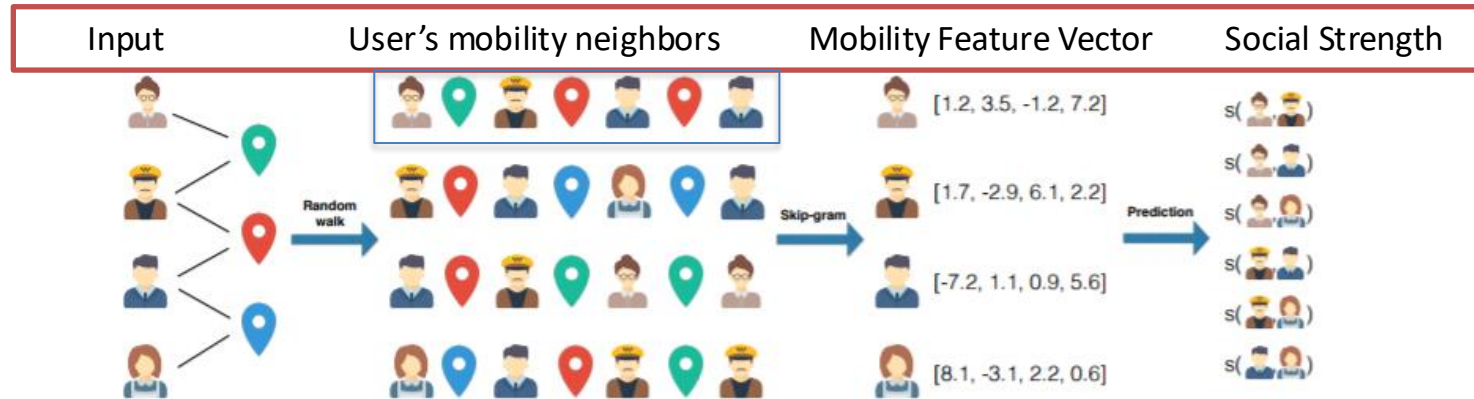


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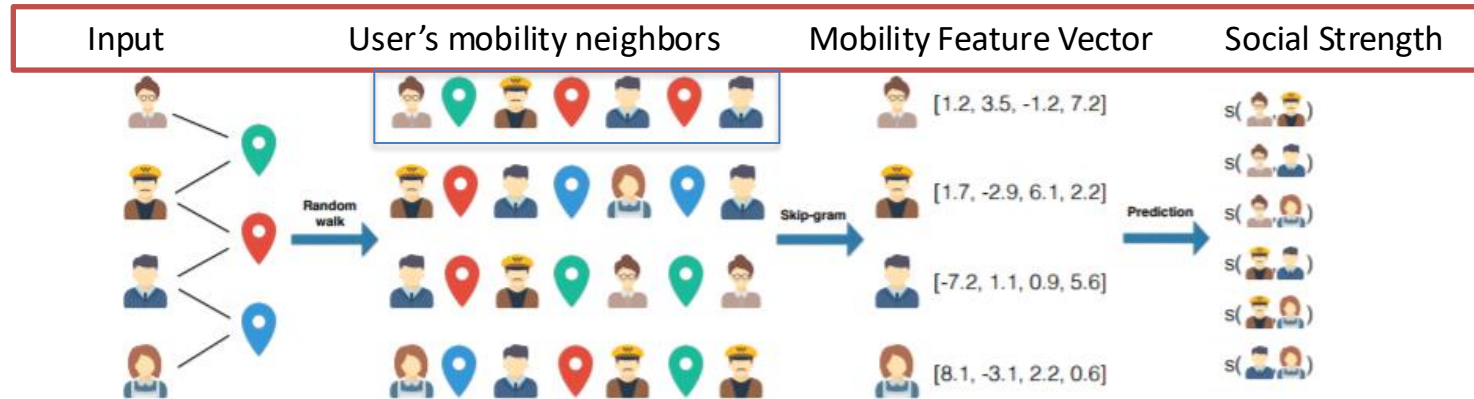
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- ✓ Performs ~10-15% percent better than EBM on relatively dense datasets.
- ✗ 3-5% worse on sparse datasets.



How to protect against social inference attack?



Co-Location Privacy Risks

1. NSA PRISM (began 2007):
Mass surveillance of location data from Google, FB, Microsoft.
2. NSA's Co-Traveler program (exposed 2013):
Identifies unknown associates of a known target.
3. Domestic prosecution facilitated by co-location information as evidence of wrongdoing. [United States v. Jones, 132 S.Ct. 945 (2012)]

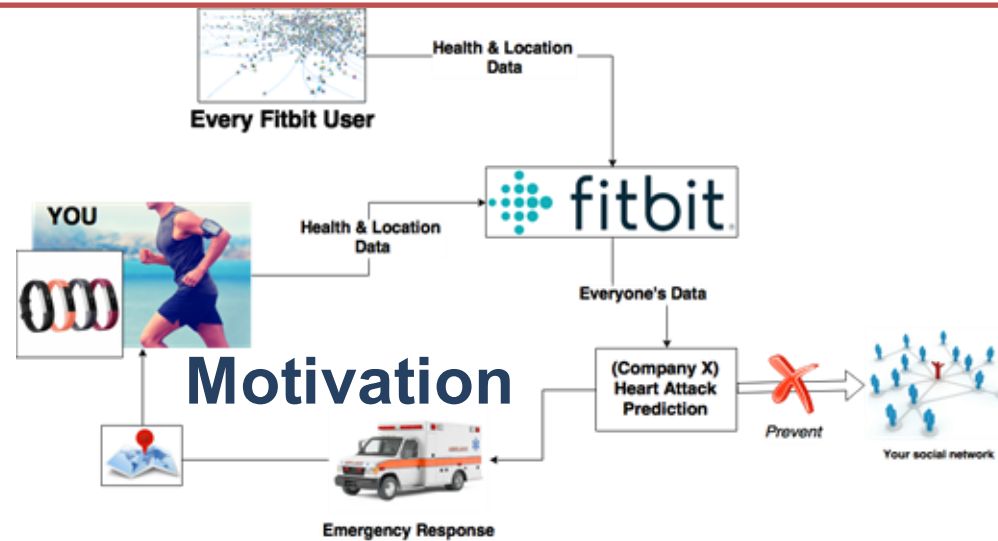


[Source: Washington Post]



Motivation

Location Data is necessary for service but social connectivity is sensitive.



Enable LBS to provide recommendation, advertisement, and other services.



Target Co-locations

The building blocks for social inference techniques.

Co-Location: Two people at *roughly* the same geographic locale at roughly the same time.



We quantify 'roughly' based on parameters Δ_s and Δ_t .

In running example,

- Assume buildings are points

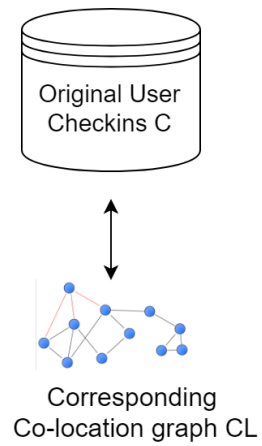
$\Delta_s = \text{SameBuilding}, \Delta_t = 1t$

Co-Locations: $(u_1, u_2), (u_5, u_6)$

Δ_s and Δ_t are application specific.

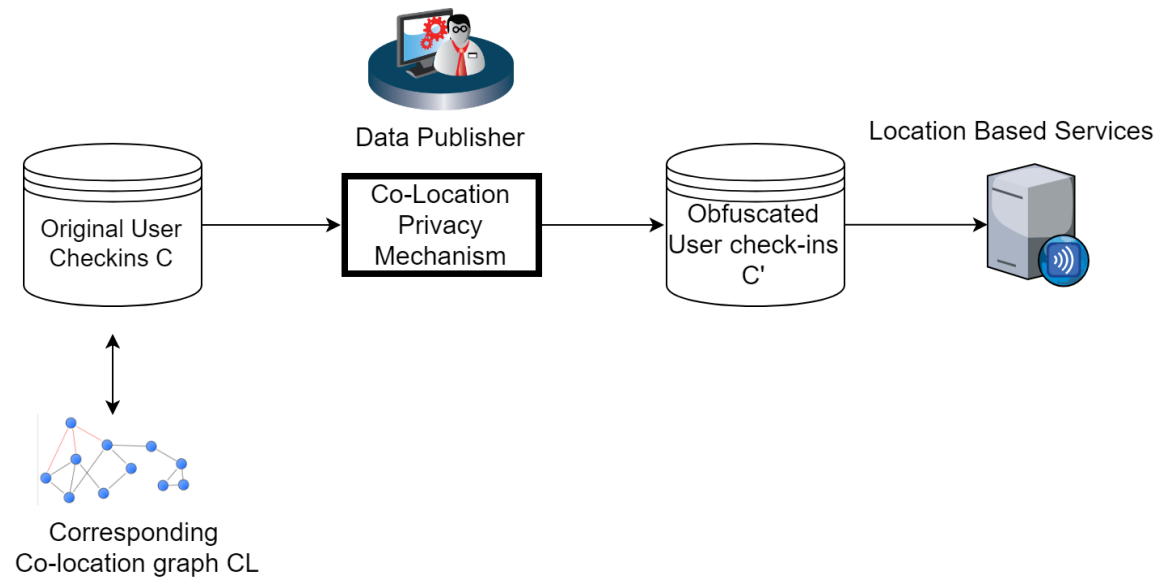


System Model



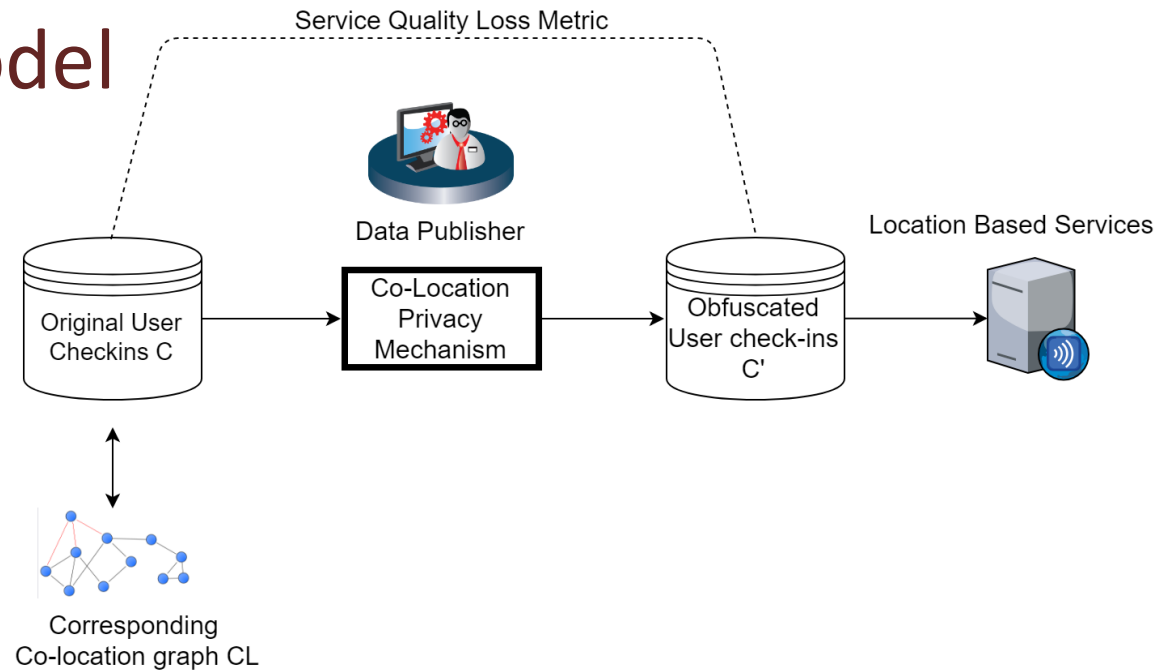


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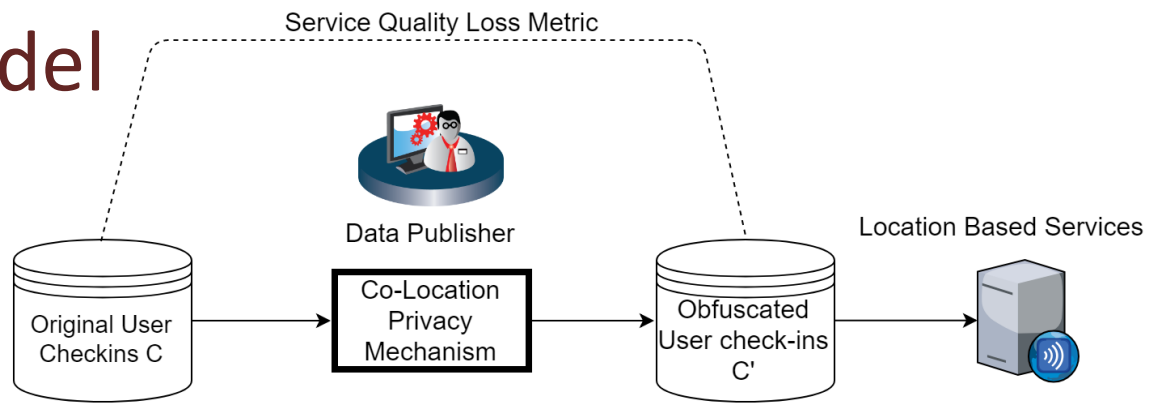


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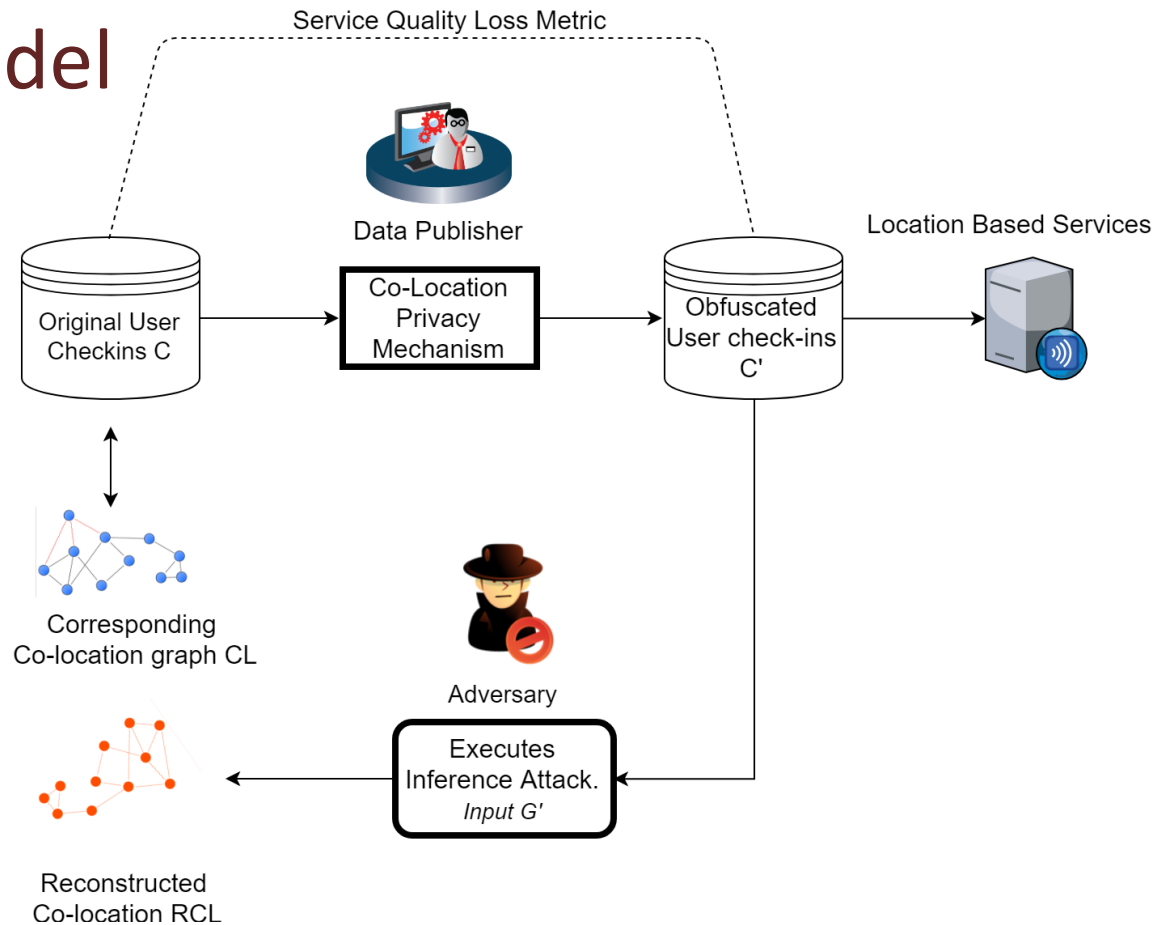
Service Quality Loss $SQL_u^i = \alpha \cdot \underbrace{\frac{\|c_u^i.l, c_u^i.l'\|}{MAX_S}}_{\text{Spatial Displacement}} + (1 - \alpha) \cdot \underbrace{\frac{|c_u^i.t, c_u^i.t'|}{MAX_T}}_{\text{Temporal Distortion}}$

c_u^i : i^{th} check-in of user u
 MAX_S, MAX_T : normalizing constants.
 α : weighting factor

Perturbed location co-ordinate $c_u^i.l'$
 Perturbed timestamp $c_u^i.t'$

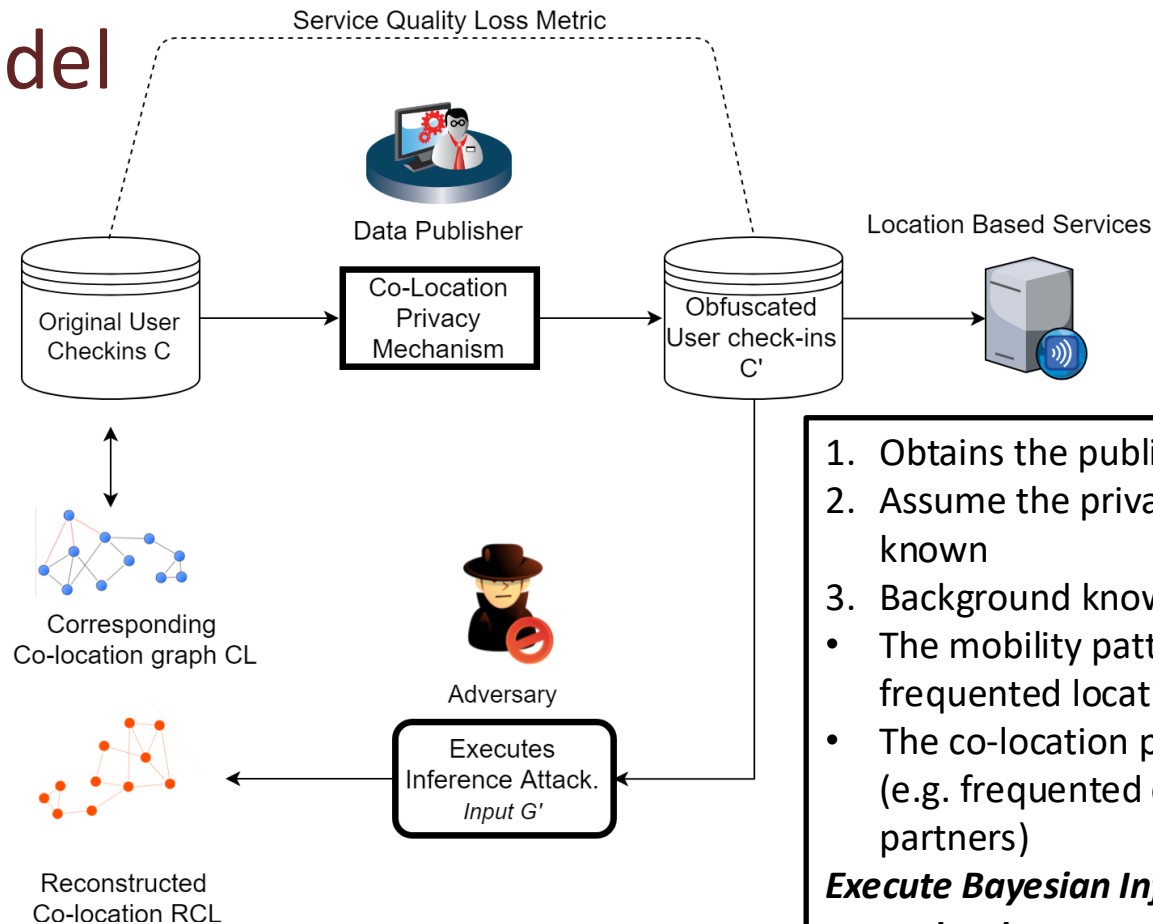


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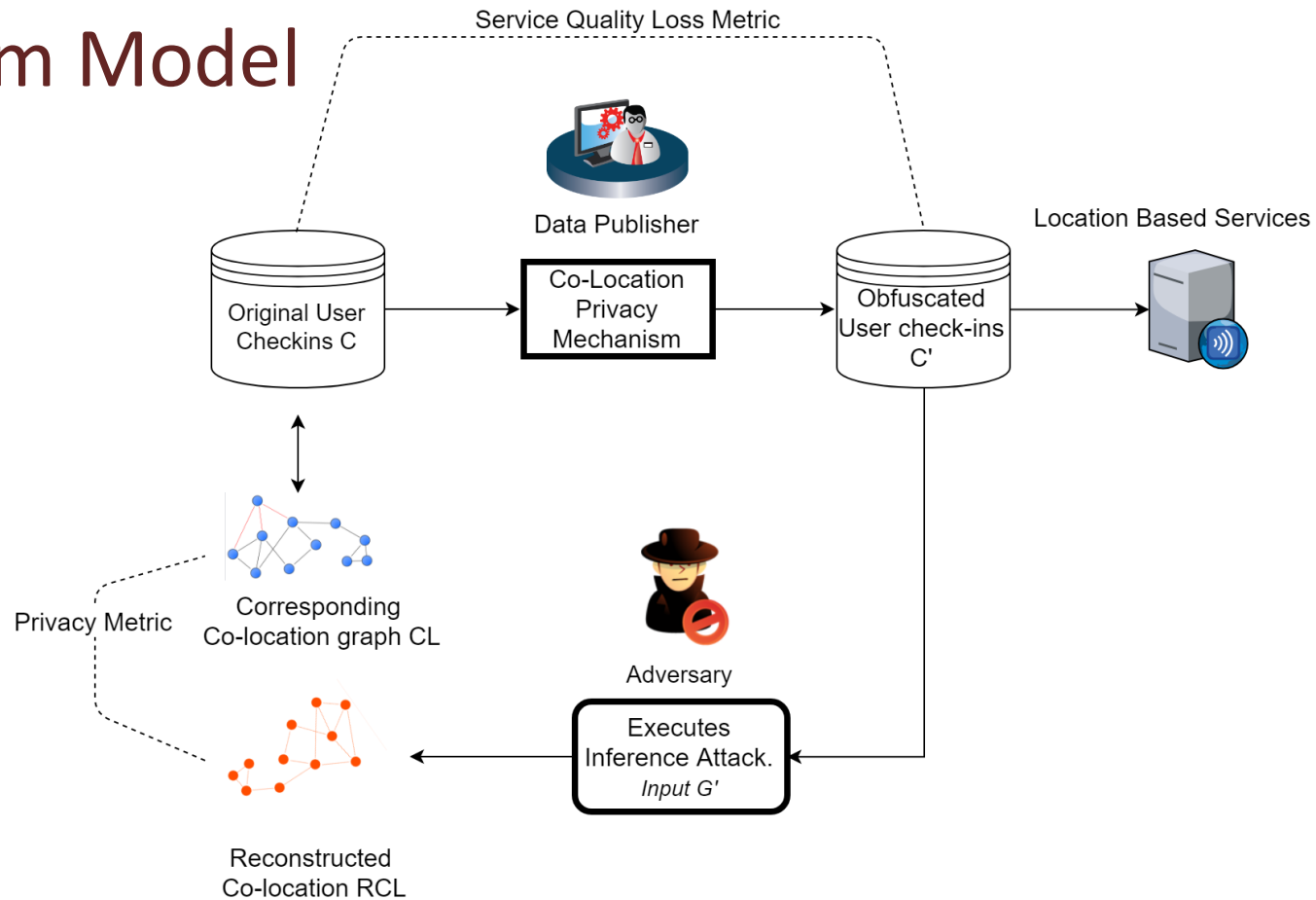


1. Obtains the published noisy data
2. Assume the privacy mechanism is known
3. Background knowledge:
 - The mobility patterns of users. (e.g. frequented locations)
 - The co-location patterns of users. (e.g. frequented co-locating partners)

Execute Bayesian Inference to reconstruct as accurate as possible representation of the original co-locations.

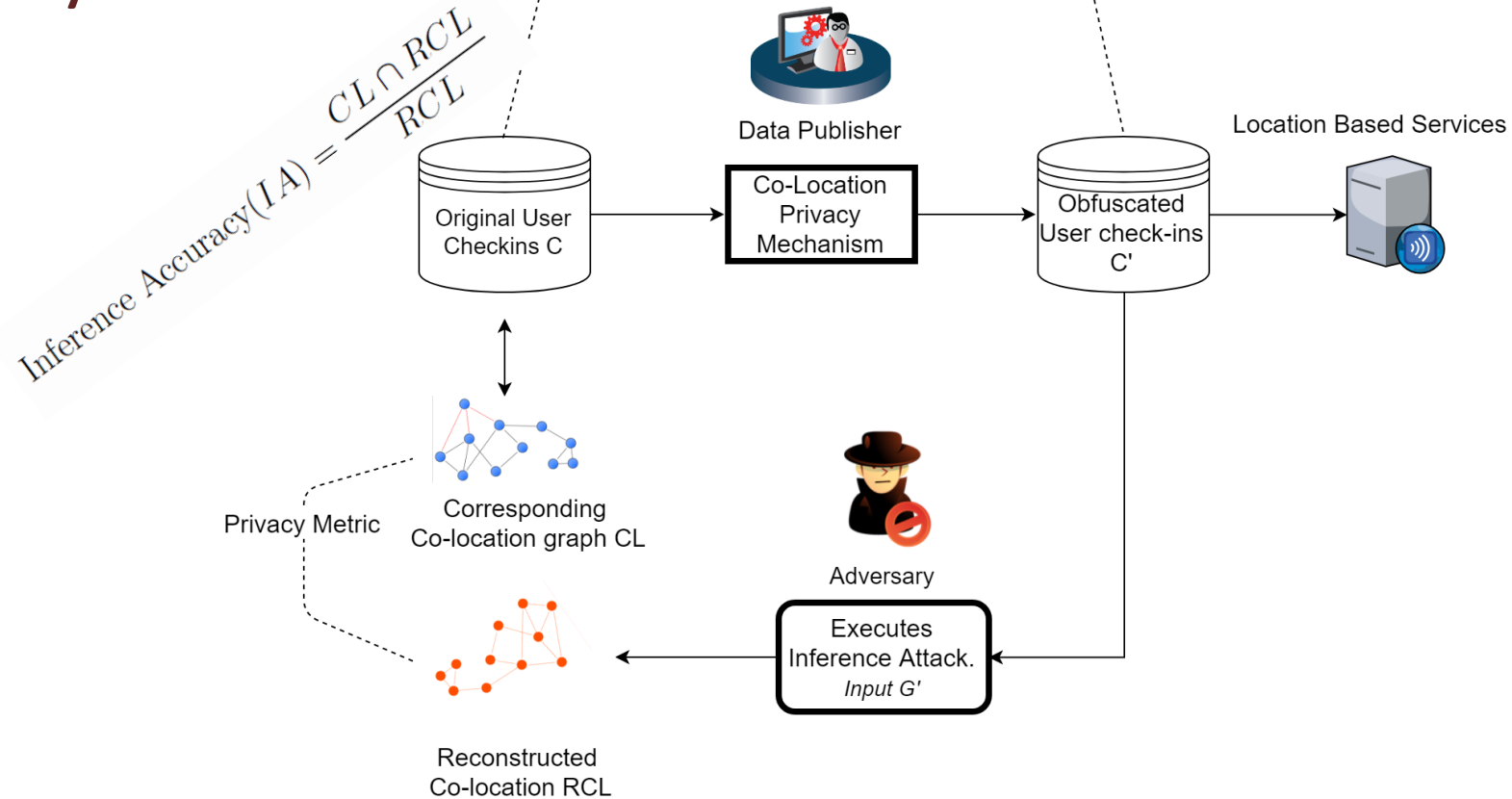


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Co-Location Privacy Mechanism 1: Gaussian Perturbation (Naïve)



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Most popular methods in statistical data privacy.

Simplest method in Location Privacy and a mechanism of noise for advanced methods like *probabilistic differential privacy*.



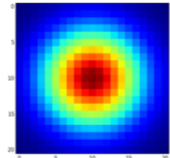
Co-Location Privacy Mechanism 1: Gaussian Perturbation (Naïve)

Most popular methods in statistical data privacy.

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Method:

1. For every co-location, it is enough to perturb one check-in.
2. Translate both coordinates with 2d-gaussian noise.
3. Translate timestamp with 1d-gaussian noise



(a) Original GPS data



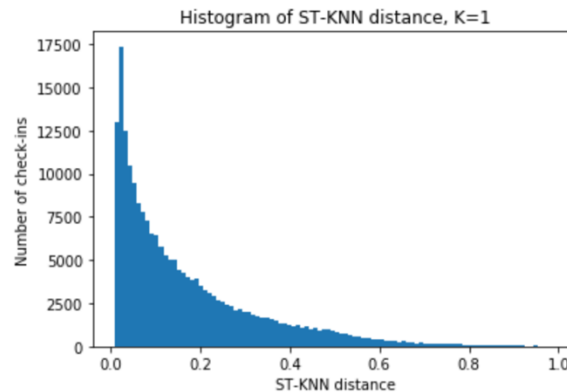
(b) Additive Gaussian noise

Krumm, [PerCom'07]



Shortcomings of Gaussian Perturbation

1. Skewed nature of the distribution of the closest neighbor:
large number of users have NN very close, while some have their NN very far.



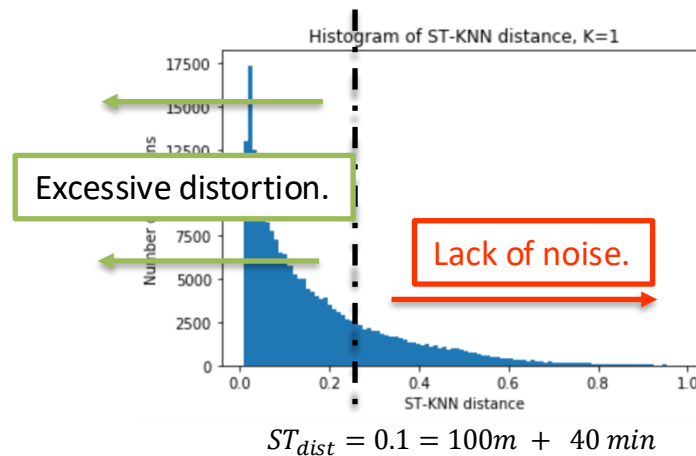
$$ST_{dist} = 0.1 = 100m + 40 \text{ min}$$

On X-Axis, 0.01 is the first 1% percent of **co-locations** (i.e. the 1st percentile) with the smallest STdist to their nearest neighbor.



Shortcomings of Gaussian Perturbation

1. Skewed nature of the distribution of the closest neighbor:
large number of users have NN very close, while some have their NN very far.
2. Any fixed magnitude of noise will lead to either:
 - **Low Privacy: Under-protected in sparse areas, or**
 - **Low Utility: Over-protected in dense areas inhibiting quality of LBSs.**



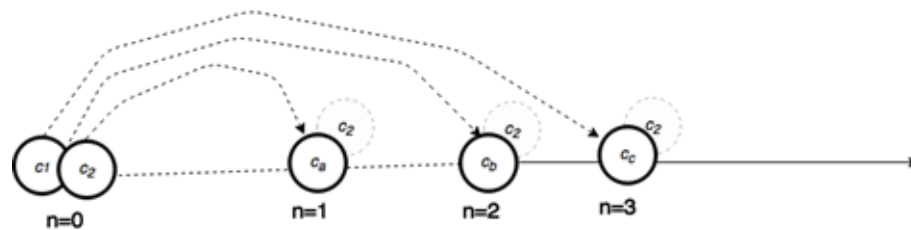
On X-Axis, 0.01 is the first 1% percent of **co-locations** (i.e. the 1st percentile) with the smallest STdist to their nearest neighbor.



Co-Location Privacy Mechanism 2: Adaptive Perturbation

Use the presence of spatiotemporal nearest neighbors as an estimate for density.

- Method:**
1. For every co-location pair, pick one check-in at random;
 2. Chose p uniformly over the set of
 - (i) the b nearest neighbors,
 - (ii) together with the current location.
 3. Move to p .



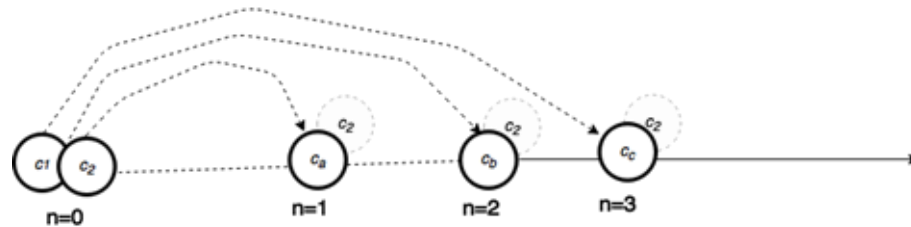
Move c_2 to any of ' $b=4$ ' positions at random

* $ST_{dist}(c, c')$ = sum of normalized spatial and temporal distances



Method:

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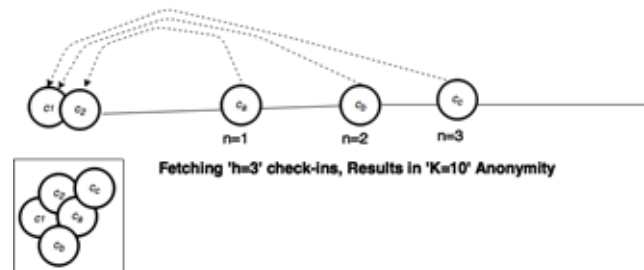
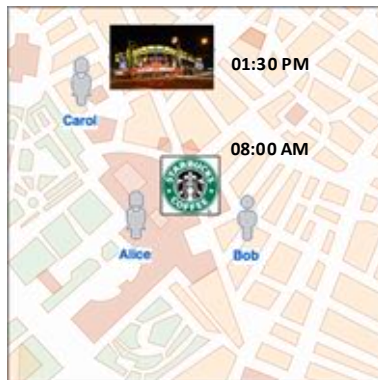


Co-Location Privacy Mechanism 3:

Co-Location K-Anonymity

Definition: A co-location is k -anonymous if it is spatiotemporally indistinguishable to $k - 1$ other co-locations.

Method: For every co-location pair, Make each co-location k -anonymous by moving “ h ” closest check-ins to form a group.



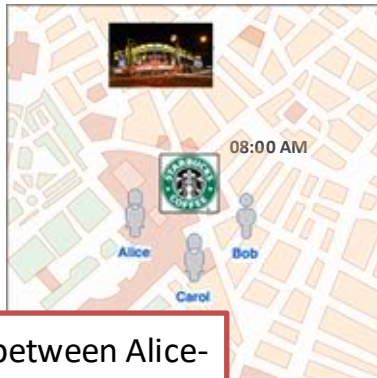


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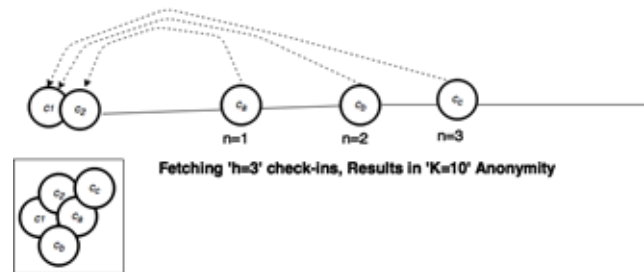
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The co-Location between Alice-Bob is now 3-anonymous.



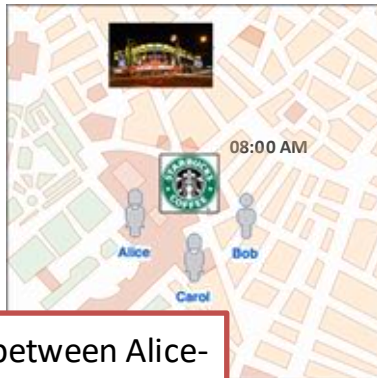


Co-Location Privacy Mechanism 3:

Co-Location K-Anonymity

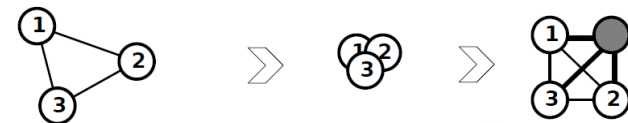
Definition: A co-location is k -anonymous if it is spatiotemporally indistinguishable to $k - 1$ other co-locations.

Method: For every co-location pair, Make each co-location k -anonymous by moving “h” closest check-ins to form a group.



The co-Location between Alice-Bob is now 3-anonymous.

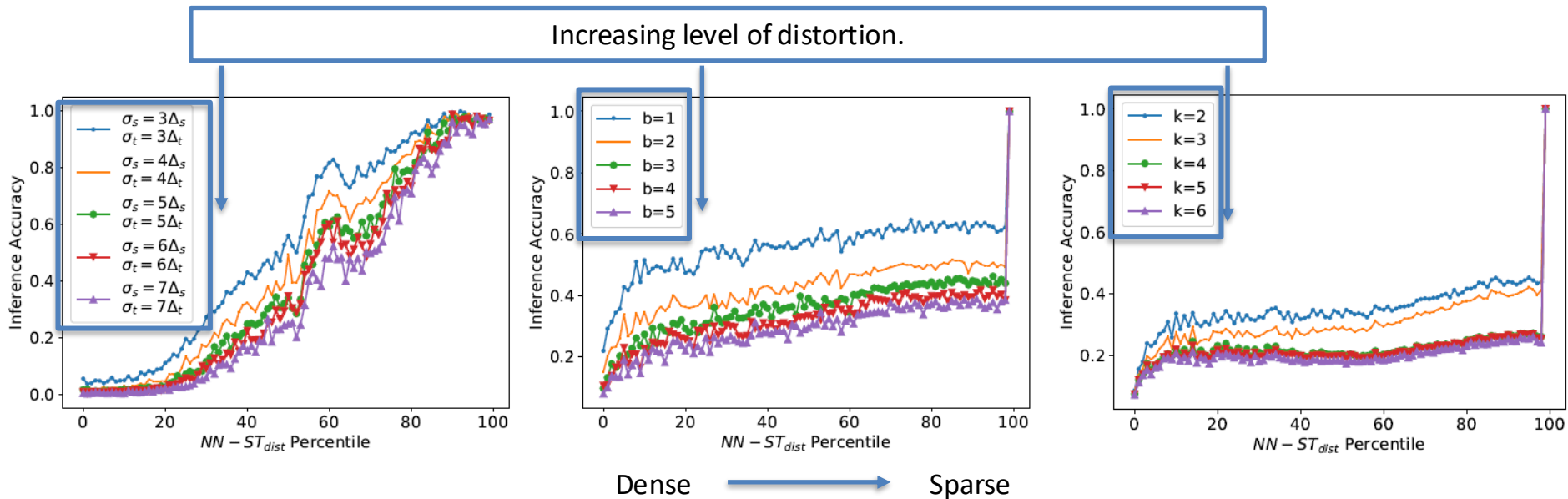
E.g. Co-location component is 2-anonymized



On seeing any co-location the adversary can only tell it's truthfulness with a certainty of $1/2$ (*i. e.* $1/k$).

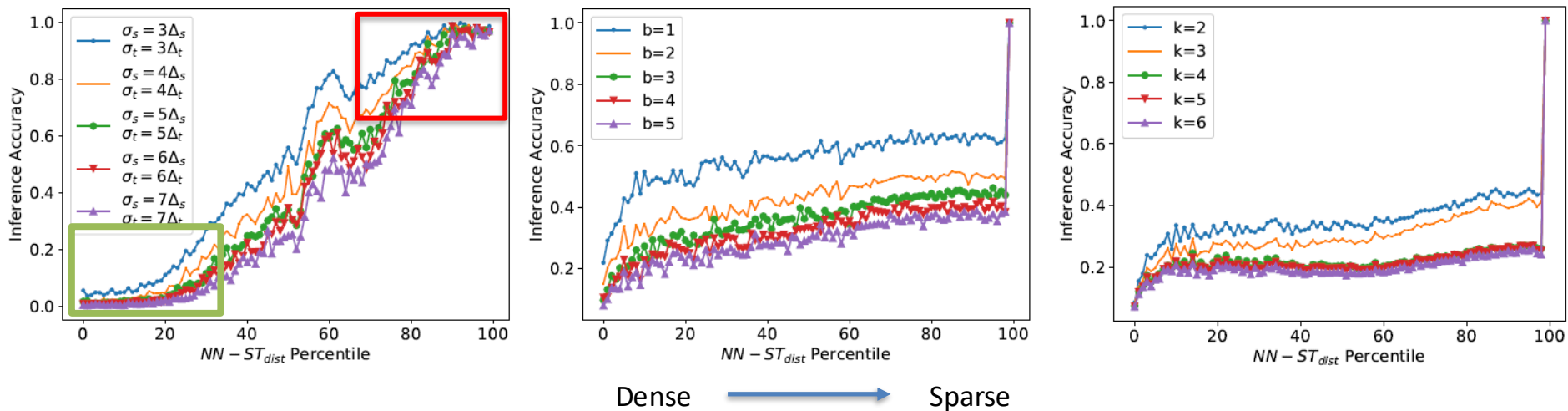


Attack Accuracy on Privacy Mechanisms





Attack Accuracy on Privacy Mechanisms

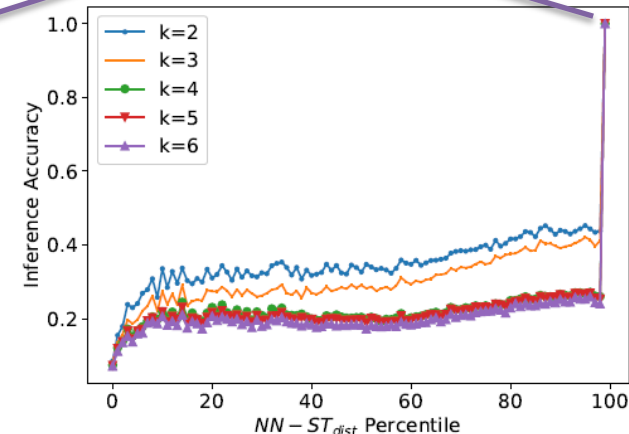
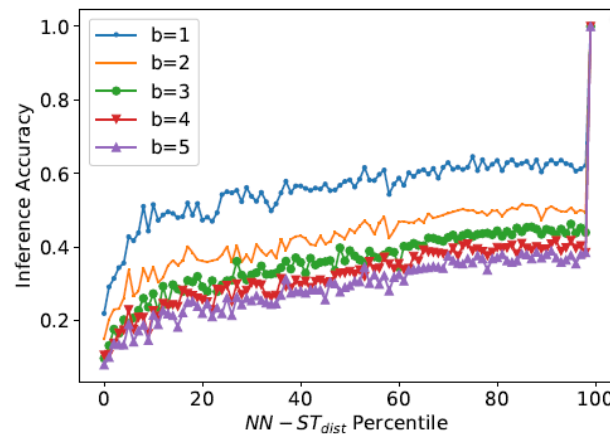
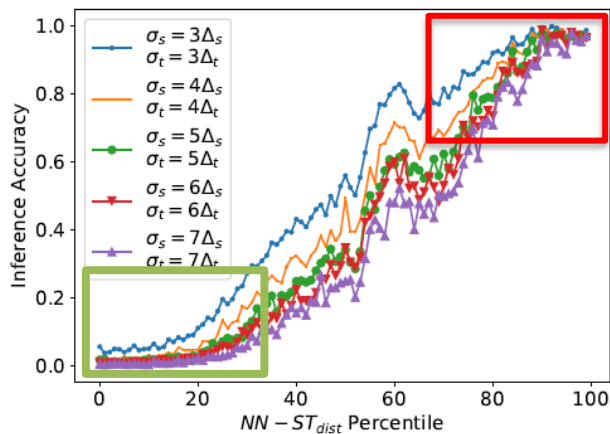


Gaussian exposes a significant portion of the population to highly accurate inferences.



Attack Accuracy on Privacy Mechanisms

Ignoring a few hundred co-locations in extremely remote locations for fair comparison.



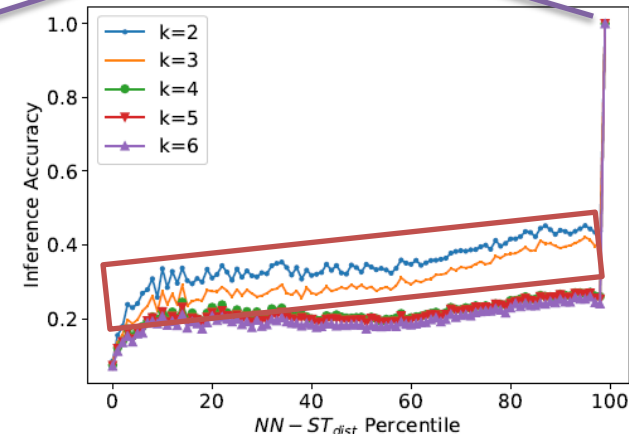
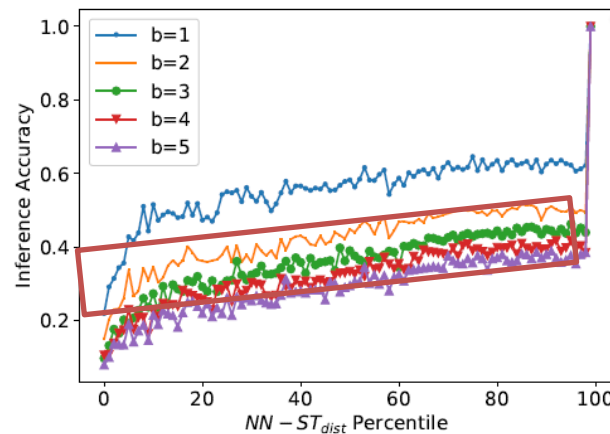
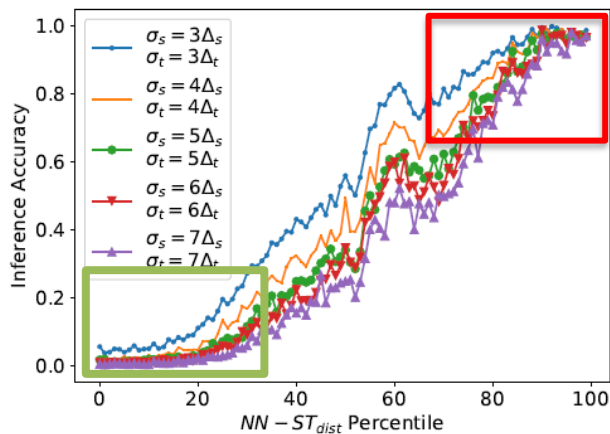
Dense $\longrightarrow$ Sparse

Gaussian exposes a significant portion of the population to highly accurate inferences.



Attack Accuracy on Privacy Mechanisms

Ignoring a few hundred co-locations in extremely remote locations for fair comparison.



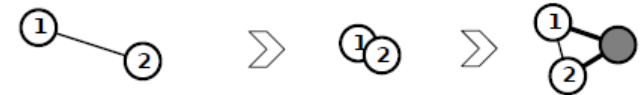
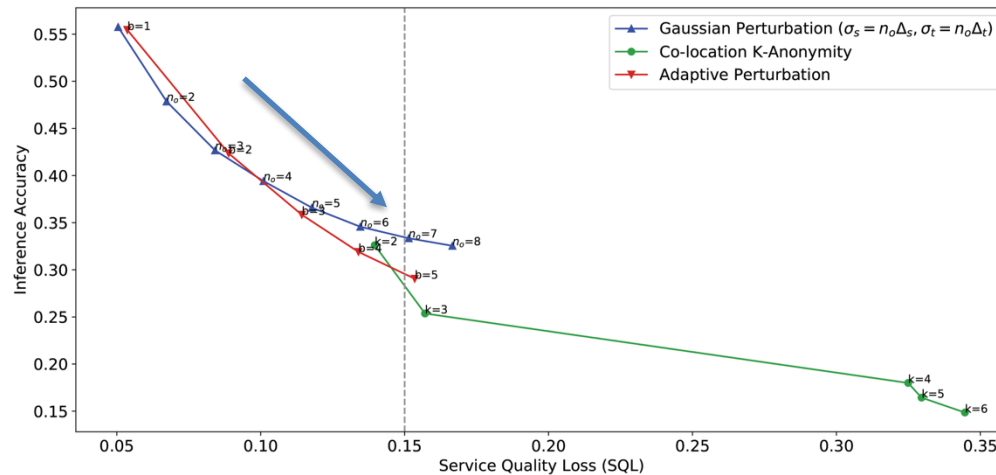
Dense $\longrightarrow$ Sparse

Gaussian exposes a significant portion of the population to highly accurate inferences.

Adaptive and k -anonymity provide consistent protection (i.e. with low variance) against an adversary.

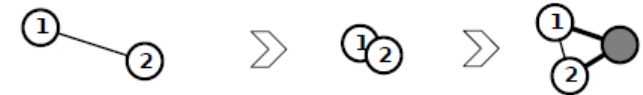
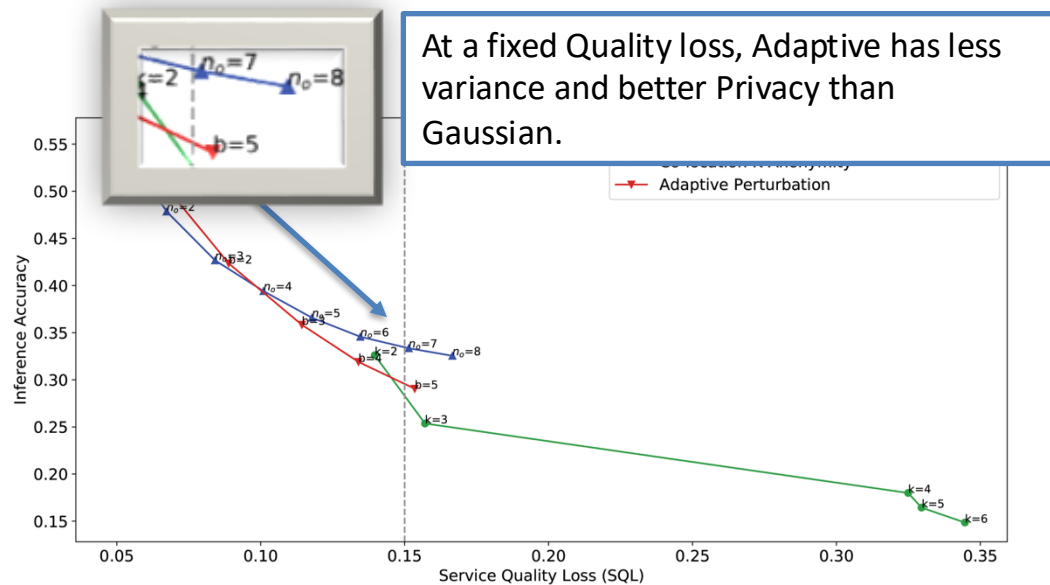


Analysis of Quality Loss





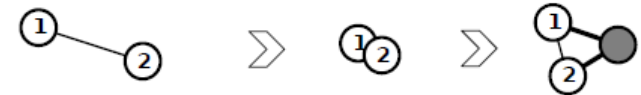
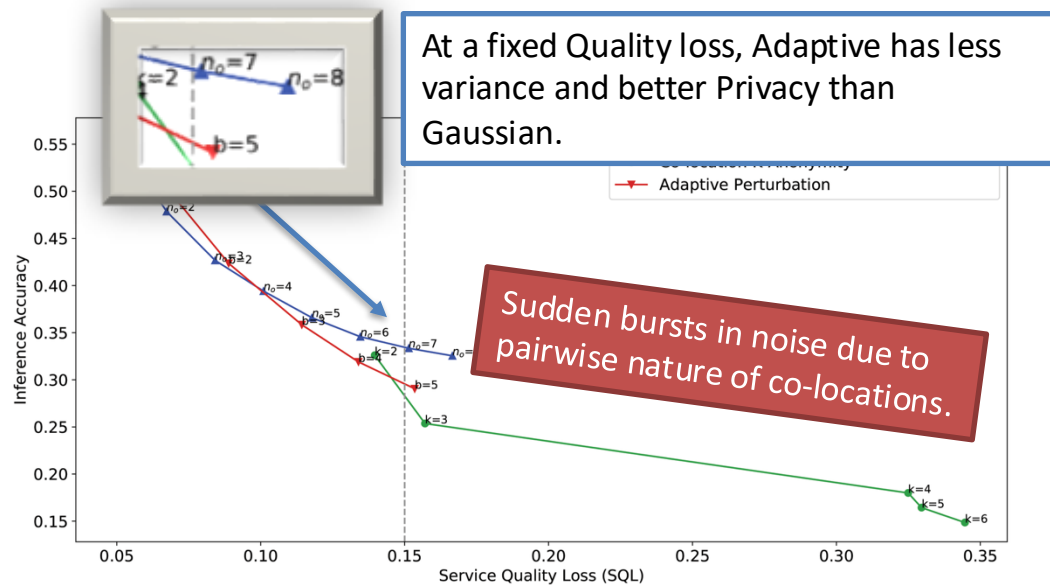
Analysis of Quality Loss



❖ Adaptive outperforms Gaussian by achieving better privacy at a given *SQL*.



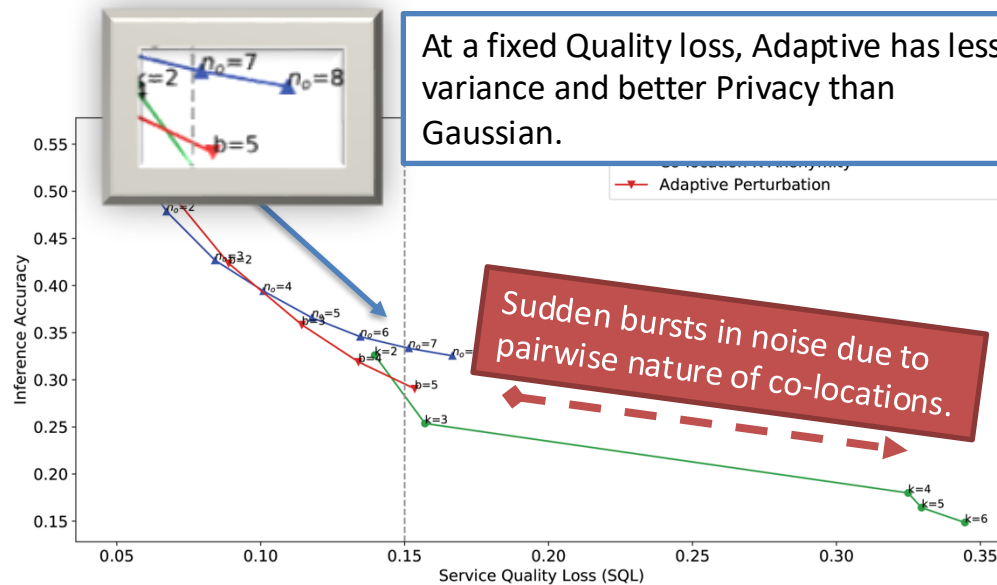
Analysis of Quality Loss



- ❖ Adaptive outperforms Gaussian by achieving better privacy at a given *SQL*.
- ❖ Co-location *k*-anonymity offers limited flexibility in calibrating noise.



Analysis of Quality Loss



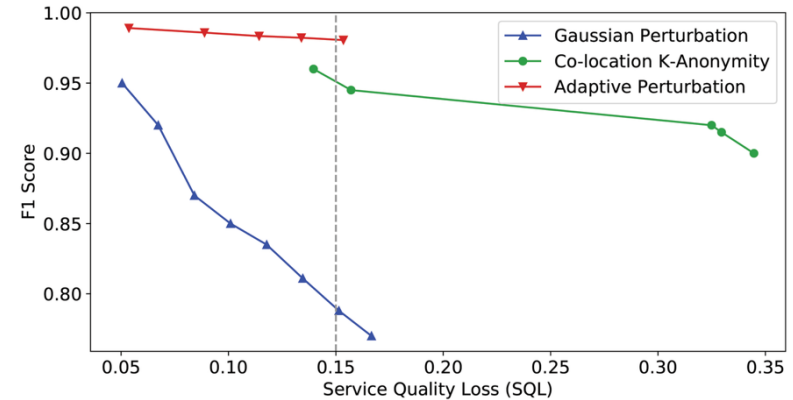
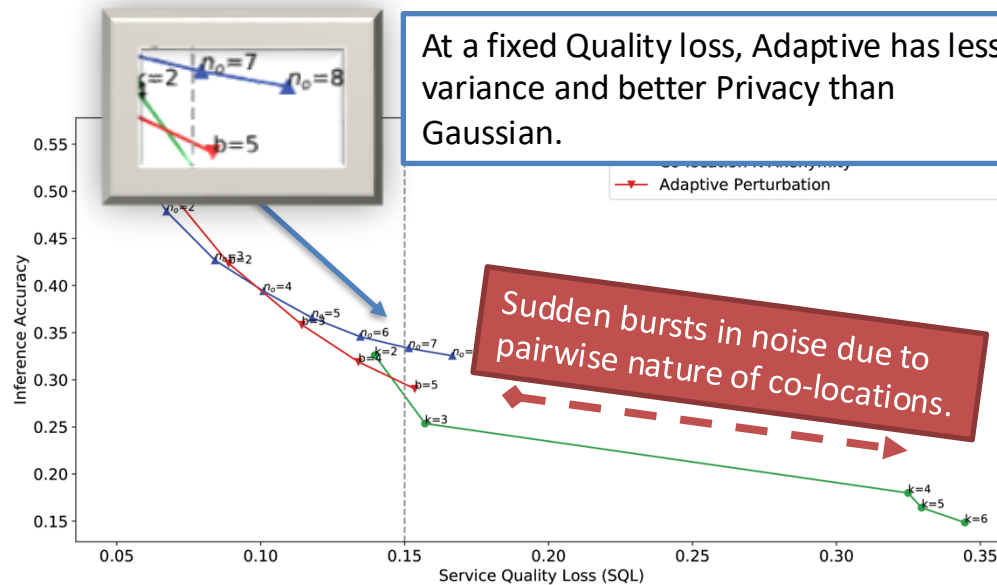
For a co-located pair 2-anonymization also 3-anonymizes.
Similarly, 4-anonymization also 5-anonymizes

BUT need to add 2 more users for 4-anonymization.

- ❖ Adaptive outperforms Gaussian by achieving better privacy at a given SQL.
- ❖ Co-location k -anonymity offers limited flexibility in calibrating noise.



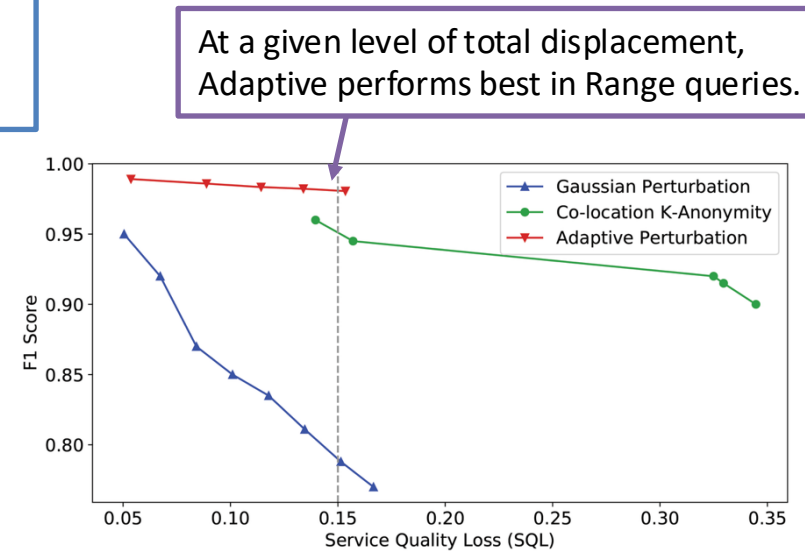
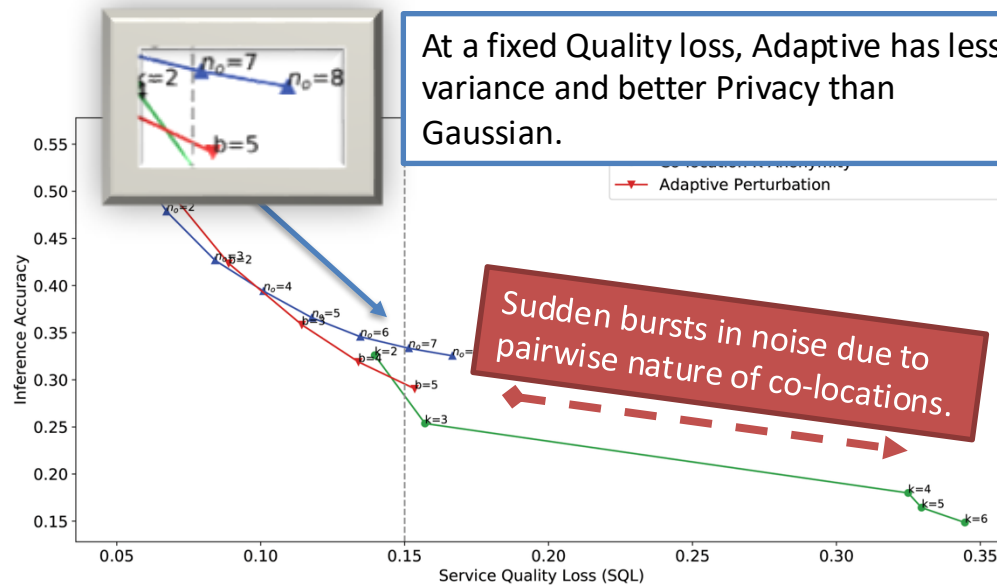
Analysis of Quality Loss and LBS Range Utility



- ❖ Adaptive outperforms Gaussian by achieving better privacy at a given *SQL*.
- ❖ Co-location *k*-anonymity offers limited flexibility in calibrating noise.



Analysis of Quality Loss and LBS Range Utility



- ❖ Adaptive outperforms Gaussian by achieving better privacy at a given *SQL*.
- ❖ Co-location *k*-anonymity offers limited flexibility in calibrating noise.
- ❖ Adaptive distorts to the NNs, hence is ideal for location-based advertising.



Thanks!